Supplementary Material

Circularity information readiness and demonstrated AI capability in architectural design

This supplement documents two linked but analytically distinct evidence streams: a primary circularity-information review and a targeted supplementary AI evidence review. It provides the search records, decision boundaries, normalized-attribute dictionary, provenance audit, AI evidence codebooks, paper index, and complete capability matrix needed to audit the reported findings without redistributing copyrighted full texts.

S1 Supplementary review methods

S1.1 Primary review search strategy

The primary review searched Scopus, Web of Science Core Collection, and IEEE Xplore using strict and controlled bridge searches. The archived exports are dated 10 July 2026. The exact query-execution date was not recoverable from the archived records. Searches covered 2010-2026, English-language articles, reviews, conference papers, and proceedings papers. Subject or category restrictions were to be applied only if term-based refinement remained too broad.

The strict logic combined building circularity, assessment, building/architectural context, and AI/drawing or BIM/structured-information concepts. The bridge logic omitted the assessment block to recover records that supplied eligible circularity-information or digital-information relationships without explicit metric terminology. Strict-set overlap was removed from Scopus and Web of Science bridge exports. IEEE Xplore did not support strict-set subtraction in the interface, so the complete filtered bridge set was exported.

Scopus S1 strict combined

```
TITLE-ABS-KEY( ("circular economy" OR circularity OR "building circularity" OR "circular building*" OR
"material circularity" OR "circular construction" OR "circular design" OR "design for disassembly" OR "design
for deconstruction" OR "design for adaptability" OR "material passport*" OR "building passport*" OR "digital
product passport*" OR "reuse potential" OR "material recovery") AND (indicator* OR metric* OR index OR indices
OR assess* OR framework* OR score* OR scoring OR measur* OR "decision support" OR "circularity assessment" OR
"circularity indicator*" OR "assessment tool*") AND (building* OR construction OR "built environment" OR
architectur* OR "architectural design" OR "building design" OR "architectural drawing*" OR "construction
drawing*" OR "technical drawing*" OR "floor plan*" OR "plan drawing*" OR "section drawing*" OR elevation* OR
"building envelope*" OR facade* OR façade* OR "building component*" OR "building assembl*") AND ((("artificial
intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "image recognition" OR "object
detection" OR "semantic segmentation" OR "instance segmentation" OR OCR OR "optical character recognition")
AND ("architectural drawing*" OR "construction drawing*" OR "technical drawing*" OR "floor plan*" OR "plan
drawing*" OR "section drawing*" OR elevation* OR "drawing recognition" OR "drawing understanding" OR
"information extraction" OR "feature extraction" OR "plan recognition" OR "symbol detection" OR "room
detection" OR "wall detection" OR "window detection" OR "door detection" OR "plan-to-BIM" OR "drawing-to-BIM"
OR "automated extraction")) OR (BIM OR "building information model*" OR IFC OR "industry foundation classes"
OR "data model*" OR ontolog* OR "knowledge graph*" OR "machine-readable" OR "machine readable" OR
"computational design" OR "parametric design" OR "generative design" OR "design automation" OR "automated
assessment")) )
```

Filters. Years 2010–2026; English; Article, Review, Conference Paper / Proceedings Paper. Apply subject/category filters only if term-based refinement remains too large.

Counts. 554 raw; 462 filtered; 462 exported.

Export. Scopus_New_S1_Strict.csv

Scopus S2 controlled bridge

```
TITLE-ABS-KEY( ("circular economy" OR circularity OR "building circularity" OR "circular building*" OR
"material circularity" OR "circular construction" OR "circular design" OR "design for disassembly" OR "design
for deconstruction" OR "design for adaptability" OR "material passport*" OR "building passport*" OR "digital
product passport*" OR "reuse potential" OR "material recovery") AND (building* OR construction OR "built
environment" OR architectur* OR "architectural design" OR "building design" OR "architectural drawing*" OR
"construction drawing*" OR "technical drawing*" OR "floor plan*" OR "plan drawing*" OR "section drawing*" OR
elevation* OR "building envelope*" OR facade* OR façade* OR "building component*" OR "building assembl*") AND
((("artificial intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "image
```

```
recognition" OR "object detection" OR "semantic segmentation" OR "instance segmentation" OR OCR OR "optical
character recognition") AND ("architectural drawing*" OR "construction drawing*" OR "technical drawing*" OR
"floor plan*" OR "plan drawing*" OR "section drawing*" OR elevation* OR "drawing recognition" OR "drawing
understanding" OR "information extraction" OR "feature extraction" OR "plan recognition" OR "symbol detection"
OR "room detection" OR "wall detection" OR "window detection" OR "door detection" OR "plan-to-BIM" OR
"drawing-to-BIM" OR "automated extraction")) OR (BIM OR "building information model*" OR IFC OR "industry
foundation classes" OR "data model*" OR ontolog* OR "knowledge graph*" OR "machine-readable" OR "machine
readable" OR "computational design" OR "parametric design" OR "generative design" OR "design automation" OR
"automated assessment")) )
```

Filters. Years 2010–2026; English; Article, Review, Conference Paper / Proceedings Paper. Apply subject/category filters only if term-based refinement remains too large.

Counts. 830 raw; 679 filtered; 217 additional records exported after removing S1 overlap.

Export. Scopus_New_S2_Bridge.csv

Web of Science W1 strict combined

```
TS=("circular economy" OR circularity OR "building circularity" OR "circular building*" OR "material
circularity" OR "circular construction" OR "circular design" OR "design for disassembly" OR "design for
deconstruction" OR "design for adaptability" OR "material passport*" OR "building passport*" OR "digital
product passport*" OR "reuse potential" OR "material recovery") AND TS=(indicator* OR metric* OR index OR
indices OR assess* OR framework* OR score* OR scoring OR measur* OR "decision support" OR "circularity
assessment" OR "circularity indicator*" OR "assessment tool*") AND TS=(building* OR construction OR "built
environment" OR architectur* OR "architectural design" OR "building design" OR "architectural drawing*" OR
"construction drawing*" OR "technical drawing*" OR "floor plan*" OR "plan drawing*" OR "section drawing*" OR
elevation* OR "building envelope*" OR facade* OR façade* OR "building component*" OR "building assembl*") AND
TS=(("artificial intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "image
recognition" OR "object detection" OR "semantic segmentation" OR "instance segmentation" OR OCR OR "optical
character recognition") AND ("architectural drawing*" OR "construction drawing*" OR "technical drawing*" OR
"floor plan*" OR "plan drawing*" OR "section drawing*" OR elevation* OR "drawing recognition" OR "drawing
understanding" OR "information extraction" OR "feature extraction" OR "plan recognition" OR "symbol detection"
OR "room detection" OR "wall detection" OR "window detection" OR "door detection" OR "plan-to-BIM" OR
"drawing-to-BIM" OR "automated extraction")) OR (BIM OR "building information model*" OR IFC OR "industry
foundation classes" OR "data model*" OR ontolog* OR "knowledge graph*" OR "machine-readable" OR "machine
readable" OR "computational design" OR "parametric design" OR "generative design" OR "design automation" OR
"automated assessment"))
```

Filters. Years 2010–2026; English; Article, Review, Conference Paper / Proceedings Paper. Apply subject/category filters only if term-based refinement remains too large.

Counts. 414 raw; 407 filtered; 407 exported.

Export. WoS_New_W1_Strict.csv

Web of Science W2 controlled bridge

```
TS=("circular economy" OR circularity OR "building circularity" OR "circular building*" OR "material
circularity" OR "circular construction" OR "circular design" OR "design for disassembly" OR "design for
deconstruction" OR "design for adaptability" OR "material passport*" OR "building passport*" OR "digital
product passport*" OR "reuse potential" OR "material recovery") AND TS=(building* OR construction OR "built
environment" OR architectur* OR "architectural design" OR "building design" OR "architectural drawing*" OR
"construction drawing*" OR "technical drawing*" OR "floor plan*" OR "plan drawing*" OR "section drawing*" OR
elevation* OR "building envelope*" OR facade* OR façade* OR "building component*" OR "building assembl*") AND
TS=(("artificial intelligence" OR "machine learning" OR "deep learning" OR "computer vision" OR "image
recognition" OR "object detection" OR "semantic segmentation" OR "instance segmentation" OR OCR OR "optical
character recognition") AND ("architectural drawing*" OR "construction drawing*" OR "technical drawing*" OR
"floor plan*" OR "plan drawing*" OR "section drawing*" OR elevation* OR "drawing recognition" OR "drawing
understanding" OR "information extraction" OR "feature extraction" OR "plan recognition" OR "symbol detection"
OR "room detection" OR "wall detection" OR "window detection" OR "door detection" OR "plan-to-BIM" OR
"drawing-to-BIM" OR "automated extraction")) OR (BIM OR "building information model*" OR IFC OR "industry
foundation classes" OR "data model*" OR ontolog* OR "knowledge graph*" OR "machine-readable" OR "machine
readable" OR "computational design" OR "parametric design" OR "generative design" OR "design automation" OR
"automated assessment"))
```

Filters. Years 2010–2026; English; Article, Review, Conference Paper / Proceedings Paper. Apply subject/category filters only if term-based refinement remains too large.

Counts. 574 raw; 558 filtered; 151 additional records exported after removing W1 overlap.

Export. WoS_New_W2_Bridge.csv

IEEE Xplore searches

IEEE Xplore I1 yielded 217 raw and 203 filtered records. I2 yielded 370 raw and 322 filtered records. The exact executed IEEE Xplore strings were not recoverable from the archived records and are therefore not reconstructed here.

S1.2 Targeted AI evidence search

A targeted Scopus search investigated capability gaps identified through the primary review. Eleven finalized search families addressed direct circularity AI, design-representation extraction, external-metadata extraction, inference/completion, carbon and life-cycle assessment prediction, reuse and recovery prediction, diagnosis, material recommendation, design-action recommendation, direct circular generation/optimization, and transferable architectural generation/optimization. Filters were 2014-2026, English, and Article, Review, or Conference Paper, with no subject-area restriction. The archived log records 30 September 2026 as its plan/log date; raw exports were archived on 1 October 2026, and no separate completed-query timestamp is recorded.

AI-S0 Direct AI + building circularity overlap

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "computer vision") AND (building* OR architectur* OR "built environment") AND
("circular economy" OR "building circularity" OR "circular construction" OR "circular design" OR "design for
disassembly" OR "design for deconstruction" OR "design for reuse" OR "material passport*" OR "product
passport*" OR "reclaimed material*" OR "component reuse" OR "building reuse" OR "material reuse") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 566 exported occurrences.

Export. Scopus_AI_S0_Direct.csv

AI-S1A Extract / Retrieve

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "computer vision") AND ("architectural drawing*" OR "construction drawing*" OR
"floor plan*" OR CAD OR BIM OR IFC OR "building information model*") AND ("information extraction" OR
"semantic extraction" OR "attribute extraction" OR "property extraction" OR "object detection" OR segmentation
OR "component recognition" OR "material recognition" OR "semantic enrichment" OR "BIM enrichment" OR "IFC
enrichment") AND (building* OR architectur* OR construction) )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 1072 exported occurrences.

Export. Scopus_AI_S1A_DesignRepresentation_Extract.csv

AI-S1B Extract / Retrieve from external product and material metadata

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "natural language processing" OR "document understanding") AND ("environmental
product declaration*" OR EPD OR "material passport*" OR "digital product passport*" OR "product passport*" OR
"building product*" OR "construction product*" OR "technical datasheet*" OR "product data sheet*" OR "material
database*" OR "product database*" OR "manufacturer data") AND (extract* OR retrieve* OR parse* OR structur* OR
standardiz* OR classify* OR "information extraction" OR "data extraction" OR "information retrieval" OR
"semantic mapping") AND (building* OR construction OR architectur* OR "built environment") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 140 exported occurrences.

Export. Scopus_AI_S1B_ExternalMetadata_Extract.csv

AI-S2 Infer / Complete

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "natural language processing" OR "knowledge graph*") AND ("architectural design" OR
"building design" OR BIM OR IFC OR "building information model*" OR "architectural drawing*" OR "construction
drawing*") AND ("semantic enrichment" OR "BIM enrichment" OR "IFC enrichment" OR "information completion" OR
"data completion" OR "missing data imputation" OR "missing information" OR "attribute prediction" OR "property
prediction" OR "relationship inference" OR "component inference" OR "material inference" OR "knowledge graph
completion") AND (material* OR component* OR assembl* OR connection* OR propert* OR attribute* OR
relationship* OR semantic*) )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 82 exported occurrences.

Export. Scopus_AI_S2_InferComplete.csv

AI-S3A Predict / Assess embodied carbon and LCA-related outcomes during building design

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network*" OR
"surrogate model*" OR "gradient boosting" OR "random forest") AND ("early design" OR "conceptual design" OR
"schematic design" OR "building design" OR "architectural design") AND (predict* OR estimat* OR assess*) AND
("embodied carbon" OR "whole life carbon" OR "life cycle assessment" OR LCA) )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 128 exported occurrences.

Export. Scopus_AI_S3A_CarbonLCA_Predict.csv

AI-S3B Predict / Assess reuse, reusability, recovery, recycling, deconstruction, and disassembly outcomes

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "neural network*" OR
"surrogate model*" OR "random forest" OR "gradient boosting") AND (building* OR architectur* OR construction
OR "building component*" OR "building material*") AND (predict* OR estimat* OR assess*) AND ("reuse potential"
OR reusab* OR "recycling potential" OR recyclab* OR recoverab* OR "recovery potential" OR "material recovery"
OR "deconstruction potential" OR "disassembly potential" OR "technical reusability" OR "component reuse") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 623 exported occurrences.

Export. Scopus_AI_S3B_ReuseRecovery_Predict.csv

AI-S4 Diagnose drivers and contributing factors of circularity-related building outcomes using explainable/interpretable AI

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "explainable AI" OR XAI
OR "explainable machine learning" OR "interpretable machine learning") AND (building* OR architectur* OR
construction OR "building design" OR "architectural design") AND (diagnos* OR "feature importance" OR "feature
attribution" OR SHAP OR "SHAP value*" OR "model interpretation" OR "model explainability" OR "explainable
prediction") AND ("embodied carbon" OR "whole life carbon" OR "life cycle assessment" OR reuse OR reusab* OR
recycl* OR recyclab* OR recoverab* OR "recovery potential" OR deconstruction OR disassembly OR "reuse
potential" OR "deconstruction potential" OR "disassembly potential") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 530 exported occurrences.

Export. Scopus_AI_S4_Diagnose.csv

AI-S5A Recommend / select building materials based on circularity, carbon, reuse, recycling, or LCA criteria

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "decision support") AND ("building material*" OR "construction material*" OR
"architectural material*" OR "material selection") AND (recommend* OR "material recommendation*" OR "material
selection" OR "decision support" OR rank* OR prioritiz*) AND ("embodied carbon" OR "life cycle assessment" OR
LCA OR circular* OR reuse OR reusab* OR recycl* OR recyclab* OR "reclaimed material*" OR "design for
disassembly") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 213 exported occurrences.

Export. Scopus_AI_S5A_MaterialRecommend.csv

AI-S5B Recommend design actions, strategies, or alternatives for circularity-related building decisions

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR
"large language model*" OR "decision support" OR "intelligent decision support") AND ("architectural design"
OR "building design" OR "circular design" OR "design for disassembly" OR "design for deconstruction" OR
"design for reuse") AND ("design recommendation*" OR "design suggestion*" OR "decision support" OR recommend*
OR "alternative selection" OR "design option selection" OR "design strategy selection") AND (circular* OR
reuse OR reusab* OR disassembly OR deconstruction OR "embodied carbon" OR "reclaimed material*" OR recycl*) )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 111 exported occurrences.

Export. Scopus_AI_S5B_DesignActionRecommend.csv

AI-S6A Generate, optimize, or rank building-design alternatives using circularity, carbon, reuse, material, or disassembly criteria

```
TITLE-ABS-KEY( ("generative design" OR "generative AI" OR "parametric design" OR "evolutionary design" OR
"genetic algorithm*" OR "evolutionary algorithm*" OR "reinforcement learning" OR "multi-objective
optimization") AND ("architectural design" OR "building design" OR "building layout*" OR "building
component*") AND (generat* OR "design alternative*" OR "design option*" OR rank* OR "multi-objective
optimization" OR "design optimization") AND (circular* OR "embodied carbon" OR "whole life carbon" OR "life
cycle assessment" OR reuse OR reusab* OR "reclaimed material*" OR disassembly OR deconstruction OR recycl* OR
"material selection") )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 201 exported occurrences.

Export. Scopus_AI_S6A_CircularGenerateRank.csv

AI-S6B Transferable integrated Generate plus Rank/Optimize evidence in architectural and building design

```
TITLE-ABS-KEY( ("artificial intelligence" OR "machine learning" OR "generative AI" OR "generative design" OR
"reinforcement learning" OR "evolutionary design") AND ("architectural design" OR "building design" OR "floor
plan*" OR "building layout*" OR "space layout*") AND (generat* OR "design alternative*" OR "design option*" OR
"alternative generation") AND (rank* OR optimiz* OR "multi-objective optimization" OR "alternative selection"
OR "design selection") AND (performance OR energy OR carbon OR material* OR cost OR daylight OR thermal OR
constructability) )
```

Filters. 2014-2026; English; Article, Review, and Conference Paper; no subject-area filter.

Counts. 606 exported occurrences.

Export. Scopus_AI_S6B_TransferableGenerateRank.csv

The eleven exports contained 4,272 record occurrences in total.

S1.3 Deduplication

For the primary review, digital object identifiers (DOIs) were normalized by lowercasing and removing resolver prefixes and unnecessary punctuation. Titles were lowercased, punctuation-normalized, accent-normalized where needed, and whitespace-collapsed. Matching used DOI first and normalized title when DOI was absent or inconsistent. Strict-search records were preferred over bridge-search records, followed by metadata completeness, valid DOI, and accession data. Six exports yielded 1,762 occurrences; 674 duplicate occurrences were removed, leaving 1,088 unique records. Duplicate rows remain traceable in the archive.

For the targeted AI review, matching used normalized DOI, exact normalized title, and then normalized title plus publication year and first author. Similar but uncertain candidates were retained and marked for review rather than automatically merged. The eleven exports yielded 4,272 occurrences; 404 duplicate occurrences were removed, leaving 3,868 unique records. Multi-search provenance was retained.

S1.4 Primary title and abstract screening boundary

Include required evidence for a building-circularity information contribution, an indicator/assessment/passport relationship, or an architectural drawing, CAD, BIM, IFC, or computational-information pathway useful for cross-mapping. Maybe was used when missing or ambiguous metadata prevented a defensible decision; Maybe records advanced to retrieval. Exclude required one recorded primary reason. Reviews and conceptual papers were retained when they contributed requirements, taxonomy, framework, or source evidence and were not treated as empirical AI demonstrations solely because they discussed AI. Full operational definitions and boundary examples are in Table S1.

S1.5 Primary full-text eligibility

A readable report was included when it supplied a building/component circularity requirement, assessment, source relationship, or eligible digital-information pathway. Retrieval failure was recorded separately from substantive exclusion. Reviews and conceptual papers could support taxonomy, framework, or source claims but were not treated as empirical AI evidence without an original implemented study. Of 855 reports sought, 123 were not retrieved, 732 readable reports were assessed, one was substantively excluded, and 731 were included.

The included reports contributed to overlapping analytical views. The circularity-information view contained 726 reports and the digital-information view contained 720. These counts are not additive because the same report could supply both circularity requirements and digital-information relationships.

S1.6 Targeted AI title and abstract screening

Screening distinguished Include, Maybe, Background Review, and Exclude. A record did not need to use the word circularity if it demonstrated a transferable building-design AI function on a relevant variable or outcome. When uncertainty remained between Exclude and Maybe, the record advanced as Maybe. Reviews and framework papers were retained for background or citation chaining but were not counted as empirical capability evidence. The 3,868 unique records yielded 1,296 Include, 374 Maybe, 134 Background Review, and 2,064 Exclude decisions. Table S1 provides the AI1-AI8 definitions and boundary examples.

S1.7 Targeted AI full-text selection

Full-text processing was purposive because the objective was gap-directed capability synthesis rather than prevalence estimation. A 300-record high-priority queue emphasized direct circularity evidence and higher-agency functions. A separate transferable queue contained 1,370 records, from which 78 gap-targeted records were selected. The final empirical corpus contained 107 page-located papers. Target/Output relationships were separated from Input/Feature, Criterion, Design Variable, Constraint, Diagnostic Factor, Context Only, and Unclear roles. Direct evidence applied a function to a circularity-oriented target; Transferable evidence demonstrated the same target/function relationship in an adjacent building-design context.

S1.8 Quality checks and researcher oversight

Retained AI relationships include page and section locators. High-risk distinctions involving geometry and material inputs, life-cycle assessment criteria, connections, passports, prediction, diagnosis, recommendation, and integrated generation/ranking were rechecked against source materials during synthesis. Uncertain title/abstract records advanced to retrieval, and unresolved full-text roles were not promoted to target/output evidence.

The archived title/abstract quality-check set contained 223 rows representing 220 of 1,088 unique records (20.2%). It emphasized uncertain decisions, missing abstracts, selected exclusions, and other high-risk records. The archived full-text quality-check set contained 93 rows representing 89 of 732 readable reports (12.2%) and included provisional Maybe decisions, the substantive exclusion, unresolved mapping groups, and a deterministic sample of the principal inclusion group. Four full-text Maybe decisions changed to Include. These records do not establish dual-independent screening, universal manual confirmation, or inter-rater reliability.

Computational and language-model tools assisted preliminary screening, structured extraction, evidence organization, consistency checking, and document preparation under researcher-defined eligibility, taxonomy, provenance, function, and role rules. The archive does not establish that every record or relationship received independent manual confirmation.

Supplementary tables

Table S1. Screening and eligibility codebook.

Stream	Stage	Decision/code	Operational definition	Boundary	Example
Primary review	Title/abstract	Include	Sufficient title/abstract evidence for a circularity-information or eligible digital-information contribution.	Advance to retrieval.	A building material-passport framework or a floor-plan recognition study with a defensible mapping role.
Primary review	Title/abstract	Maybe	Plausible relevance, but building scale, method, input/output, or design relevance cannot be resolved from title/abstract.	Advance to retrieval rather than exclude.	A short BIM and circular-construction abstract with an unclear implemented method.
Primary review	Title/abstract	Exclude	The record clearly fails the documented eligibility boundary.	Assign one primary E-code.	Municipal recycling policy with no building, component, or design-information relationship.
Primary review	Title/abstract	E1	No building, construction, or architecture relationship.	Exclude only when substantive AEC context is absent.	Non-AEC industrial process study.
Primary review	Title/abstract	E2	Neither circularity nor an eligible AI/drawing-information mapping relationship.	Exclude when both evidence streams are absent.	Generic building topic with no circularity or information-extraction relevance.
Primary review	Title/abstract	E3	No indicator, assessment, framework, extraction method, or relevant tool.	Exclude when no usable analytical contribution is visible.	Descriptive commentary without an assessment or extraction contribution.
Primary review	Title/abstract	E4	No relevant AI, computational, or digital-information relationship and no eligible circularity mapping role.	Exclude only after checking both streams.	Ordinary calculation without circularity-information mapping relevance.
Primary review	Title/abstract	E5	No usable design-information or design-decision connection.	Exclude management-only material lacking information relevance.	Organizational adoption study without an attribute/source relationship.
Primary review	Title/abstract	E6	Urban, regional, policy, or macro scale only.	Exclude when building/component/system evidence cannot be separated.	City-level circularity index only.
Primary review	Title/abstract	E7	Waste management only without design, indicator, or component-recovery relevance.	Do not use when the paper links waste to building design or recovery information.	Collection logistics without a building-design information relationship.
Primary review	Title/abstract	E8	Duplicate occurrence.	Remove after DOI/title matching while retaining provenance.	Same DOI retrieved from strict and bridge searches.
Primary review	Title/abstract	E9	Inaccessible or unusable title/abstract metadata.	Use only when no meaningful screening evidence is available.	Record lacking a usable title and abstract.
Primary review	Title/abstract	E10	Other stated reason.	Record the reason explicitly.	Ineligible language or document type where documented.
Primary review	Full text	Include	Readable report supplies a building/component circularity requirement, assessment, source relationship, or eligible digital-information pathway.	Retain for the relevant analytical view(s).	Indicator method with coded inputs or a BIM extraction method that can be cross-mapped.

Stream	Stage	Decision/code	Operational definition	Boundary	Example
Primary review	Full text	FT1	Full text not retrieved.	Record separately from substantive exclusion.	Report sought but unavailable through lawful access.
Primary review	Full text	FT2	Not substantively building/AEC relevant after full-text review.	Substantive exclusion.	Manufacturing-only study without transferable building information.
Primary review	Full text	FT3-FT8	No qualifying circularity, assessment, computational-information, design, recovery, or building-scale contribution under the recorded code.	Use the specific archived reason.	Urban/macro-only report (FT8).
Primary review	Full text	FT9	Background review only and unsuitable for the claimed empirical relationship.	May remain contextual but not empirical evidence.	Broad review without an original implementation.
Primary review	Full text	FT10	Duplicate or superseded report.	Retain the more complete study representation.	Conference abstract superseded by a complete article.
Primary review	Full text	FT11	Insufficient methodological detail for computational-readiness coding.	Exclude from the affected analytical claim.	Method/output remains unresolved after reading.
Primary review	Full text	FT12	AI or drawing-extraction report with no direct circularity connection.	Eligible only where the transferability rule supports mapping; otherwise exclude from the primary claim.	Unrelated visual recognition workflow.
Primary review	Full text	FT13	Circularity metric report without direct drawing, AI, BIM, or computational connection.	May remain in the circularity-information view.	Metric definition used for attribute mapping only.
Targeted AI review	Title/abstract	Include	Credible empirical or applied evidence potentially relevant to at least one AI function and circularity/building-design attribute.	Advance to selective full-text consideration.	Early-design embodied-carbon prediction.
Targeted AI review	Title/abstract	Maybe	Potentially relevant but the AI method, target/output, AEC context, or attribute role remains uncertain.	Prefer Maybe to exclusion when uncertainty is material.	Optimization abstract that does not reveal whether alternatives are generated.
Targeted AI review	Title/abstract	Background Review	Review, framework, or perspective useful for context or citation chaining but not primary empirical evidence.	Do not count as an empirical AI relationship.	Review of digital technologies for circular construction.
Targeted AI review	Title/abstract	AI1	Not building/AEC relevant.	Requires absence of substantive AEC context.	Product optimization outside the built environment.
Targeted AI review	Title/abstract	AI2	No relevant AI/ML/intelligent computational method.	Ordinary deterministic calculation alone is insufficient.	Static spreadsheet calculation without an eligible decision-support workflow.
Targeted AI review	Title/abstract	AI3	No relevant circularity attribute or transferable building-design variable/outcome.	Do not use merely because circularity terminology is absent.	AI on a building topic unrelated to the 48 attributes or functions.
Targeted AI review	Title/abstract	AI4	Construction management/logistics only with no design-information relevance.	Exclude when no target attribute or architect-facing information relationship exists.	Crew scheduling only.
Targeted AI review	Title/abstract	AI5	Generic sustainability/performance only with no relevant target input/output.	Do not use for embodied carbon, LCA, reuse, materials, BIM/IFC, disassembly, or other explicit targets.	Thermal comfort optimization with no transferable target.

Stream	Stage	Decision/code	Operational definition	Boundary	Example
Targeted AI review	Title/abstract	AI6	Review/background only.	Normally classify as Background Review rather than empirical evidence.	Systematic review without an original implemented study.
Targeted AI review	Title/abstract	AI7	Duplicate.	Handled during deduplication.	Same DOI across search families.
Targeted AI review	Title/abstract	AI8	Insufficient usable information.	Do not use solely because an abstract is short; use Maybe if relevance remains plausible.	Unidentifiable record without usable metadata.
Targeted AI review	Full text	Include	Full text verifies an implemented target/output or clearly relevant supporting-role relationship for the targeted capability question.	Retain page/section locators.	AI extracts recycled-content fields from EPDs.
Targeted AI review	Full text	Maybe	Available full text remains genuinely ambiguous about method, target, role, or stage.	Do not promote to verified evidence.	Hybrid optimization workflow with unresolved generation step.
Targeted AI review	Full text	Background Review	Useful for context but not a primary empirical relationship.	Excluded from the 107 empirical-paper capability ledger.	Architecture-AI review.
Targeted AI review	Full text	FT-AI1 / T2-FT1	Full text not retrieved.	No evidence inferred from title/abstract.	Paywalled paper without authorized access.
Targeted AI review	Full text	FT-AI2-FT-AI10 / T2-FT2-T2-FT10	Recorded substantive reason: wrong domain, no eligible AI, no usable relationship, generic performance, management-only scope, review-only, duplicate, unclear workflow, or stated other reason.	Use one primary code and explanation.	Optimization-only paper that does not address a targeted gap.

Table S2. Ten-study evidence comparison.

Study	DOI	Source/project state	Implemented task	Target/output and result	Architectural meaning	Boundary
Akanbi et al. (2019)	10.1016/j.jclepro.2019.03.172	Revit building-design scenarios	BIM-linked whole-life and deconstruction analytics	Performance, element deconstruction, and design advice tested on three scenarios	Inspect how modeled assembly choices affect later disassembly	Weighted factors and future conditions remain modeled assumptions
Honic et al. (2021)	10.1016/j.jclepro.2021.128702	Existing building, pre-demolition audit	Passport and hazardous-material assessment	Material quantities and hazardous content documented for end-of-life action	Plan demolition, recovery, and special handling	Retrospective source; does not prove original early-design availability
Lauble et al. (2025)	10.1088/1755-1315/1546/1/012076	Existing-building documents	CV/OCR/NLP extraction and passport population	Components, geometry, volume, and material labels structured for a passport	Build an intervention inventory	Reconstruction from stock documents, not original design evidence
Vergés et al. (2026)	10.1016/j.eiar.2025.108243	8,463 product EPDs	LLM extraction to structured fields	Product, impact, composition, recycled content, service-life, transport, waste, and end-of-life data; missing fields returned as null	Compare declared product evidence	Product-specific and standard-dependent; EPD must exist

Study	DOI	Source/project state	Implemented task	Target/output and result	Architectural meaning	Boundary
Forth and De Wolf (2026)	10.1016/j.jobe.2026.115417	Three architectural BIM/IFC cases	Semantic graph enrichment	Connection and disassembly properties enriched; BCI within 1.5% of manual results	Inspect missing connection semantics during design	Low archetype-match accuracy required source review; no reversible/permanent inference
Rakhshan et al. (2021)	10.1016/j.jobe.2021.102791	Expert reuse questionnaire	Supervised reusability prediction and factor analysis	Technical reuse potential and influential component factors	Review conditions that could limit future reuse	Future condition and demand remain uncertain
Iqbal et al. (2025)	10.1016/j.mlwa.2025.100760	Preliminary reinforced-concrete dataset	Prediction, XAI, counterfactual search	Carbon, energy, and cost predictions; model attributions; predefined 10%-20% counterfactual reductions	Compare changeable quantities and logistics assumptions	Supplied data, predictive association, and bounded design variables; no product-level selection
Stellmacher et al. (2025a, 2025b)	10.35490/ec3.2025.236; 10.14455/isec.2025.12(1).con-08	EPDs and product documents	LLM extraction, knowledge graph, GraphRAG queries	Structured JSON, requirement matches, context-sensitive product responses	Retrieve evidence and compare candidate products	Related prototypes, limited independent evaluation, and product-selection timing unresolved
Xiang et al. (2024)	10.1016/j.dibe.2024.100413	Parametric prefabricated design	Integrated generation, carbon calculation, and GA optimization	Alternatives reduced overall carbon by 10.51% relative to standardization-only case	Compare formwork, element, and standardization decisions	Carbon-standardization objective is narrower than circularity and subject to model constraints
Mirzaei et al. (2025)	10.22260/isarc2025/0094	Reused-steel inventory and parametric model	Generation and NSGA-II optimization	Reused-component topology and geometry generated and optimized	Adapt structural design to available reclaimed stock	Inventory, certification, and constructability conditions govern transfer

Table S3. Complete normalized-attribute dictionary.

ID	Family	Attribute	Information meaning	Original terminology	Frequency n	Readiness	Provenance basis	Notes
ATT-01	Connections and interfaces	accessibility	Physical or visual access needed to inspect, service, remove, or separate a component.	accessibility	316	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-02	Design adaptability and standardization	adaptability	Capacity of a building, system, or component to accommodate future change.	adaptability; Adaptability/flexibility	251	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-03	Geometry and spatial definition	area	Plan, surface, gross, net, or component area used in a circularity calculation.	area; Area and volume	482	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-04	Component and assembly definition	assembly sequence	Order in which components are assembled, removed, or disassembled.	assembly sequence; Disassembly/deconstruction potential	43	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-05	Quantity and mass	component count	Number of components or repeated elements required by an assessment.	component count; Material and component quantities	103	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.

ID	Family	Attribute	Information meaning	Original terminology	Frequency n	Readiness	Provenance basis	Notes
ATT-06	Component and assembly definition	component identity	Generic or specific identification of a building component or element type.	component identity; Component identity/type	45	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-07	Component and assembly definition	component relationships	Adjacency, containment, support, connectivity, or assembly relationships among components.	component relationships; Spatial/component relationships	476	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-08	Connections and interfaces	connection type	Generic or detailed means by which components are joined.	connection type; Connection type/fasteners	363	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-09	Disassembly and deconstruction	deconstruction potential	Potential for planned dismantling or selective deconstruction rather than destructive demolition.	deconstruction potential; Disassembly/deconstruction potential	312	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-10	Geometry and spatial definition	dimensions	Linear dimensions, thicknesses, spans, or sizes used in a circularity calculation.	dimensions; Geometry and dimensions	532	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-11	Disassembly and deconstruction	disassembly potential	Potential to separate components without destructive loss of function or value.	disassembly potential; Disassembly/deconstruction potential	193	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-12	Supply, take-back, and market conditions	economic value	Current cost, residual value, or economic quantity used in a circularity assessment.	economic value; Economic/market value	557	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-13	Environmental and resource flows	embodied carbon	Embodied greenhouse-gas impact associated with materials, products, or assemblies.	embodied carbon; LCA/embodied impacts	362	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-14	Scenario and expert inputs	expert judgment	Human interpretation, weighting, or decision input required by an assessment.	expert judgment; Scenario/expert assumptions	250	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.
ATT-15	Design adaptability and standardization	flexibility	Capacity to support alternative layouts, uses, or configurations.	flexibility; Adaptability/flexibility	281	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-16	Supply, take-back, and market conditions	future value	Estimated value of a material or component at a future recovery or reuse time.	future value; Economic/market value	22	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.

ID	Family	Attribute	Information meaning	Original terminology	Frequency n	Readiness	Provenance basis	Notes
ATT-17	Geometry and spatial definition	geometry	Spatial form, shape, topology, or geometric representation of building elements.	geometry; Geometry and dimensions	370	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-18	Environmental and resource flows	LCA Data	Inventory, characterization factors, or other data used in life-cycle assessment.	life-cycle assessment data; LCA/embodied impacts	377	Core	Core: direct stage evidence	At least two independent, human-verified papers provide P1 evidence of design-stage use and provenance.
ATT-19	Geometry and spatial definition	location	Site, spatial position, orientation, or element location relevant to an assessment.	location; Geometry and dimensions	417	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-20	Service, maintenance, and replacement	maintainability	Ease and feasibility of inspection, servicing, and maintenance.	maintainability; Repair/maintenance/replaceability	19	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-21	Product and provenance metadata	manufacturer information	Manufacturer-specific product identity, properties, documentation, and service information.	manufacturer information; Manufacturer/product data	165	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-22	Quantity and mass	mass	Material or component mass used in material-flow or impact calculations.	mass; Material and component quantities	414	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-23	Product and provenance metadata	material passport	Structured material-level identity, composition, provenance, and circularity record.	material passport; EPD/passport documentation	122	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-24	Quantity and mass	material quantity	Quantity of a material by volume, mass, area, or another measure.	material quantity; Material and component quantities	247	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-25	Material identity and composition	material toxicity	Hazardous substance, toxicity, or health-related material property.	material toxicity; Toxicity/hazardous content	153	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-26	Material identity and composition	material type	Generic material family, specific material, or product-level material identity.	material type; Material identity/type	116	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-27	Connections and interfaces	mechanical fastener	Presence and type of bolts, screws, clips, or other mechanical fasteners.	mechanical fastener; Connection type/fasteners	37	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.

ID	Family	Attribute	Information meaning	Original terminology	Frequency n	Readiness	Provenance basis	Notes
ATT-28	Design adaptability and standardization	modularity	Use of coordinated modules, grids, kits, or repeatable dimensional units.	modularity; Modularity/standardization	304	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-29	Connections and interfaces	permanent connection	Presence of irreversible or difficult-to-separate joining methods.	permanent connection; Connection type/fasteners	35	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-30	Product and provenance metadata	product passport	Structured product-level identity, properties, provenance, and circularity record.	product passport; EPD/passport documentation	61	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-31	Reuse, recycling, and recovery	recovery potential	Potential for recovery through reuse, remanufacture, recycling, or other pathways.	recovery potential; Recyclability/recovery potential	88	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.
ATT-32	Material identity and composition	recycled content	Share of recycled feedstock in a material or product.	recycled content; Circular material content	38	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-33	Reuse, recycling, and recovery	recycling potential	Potential for a material or component to be recycled under specified conditions.	recycling potential; Recyclability/recovery potential	288	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.
ATT-34	Material identity and composition	renewable content	Share of renewable or bio-based content in a material or product.	renewable content; Circular material content	53	non-Core	Requirement known, provenance weak	The attribute is repeatedly required, but implementation/provenance evidence is weak or absent.
ATT-35	Service, maintenance, and replacement	repairability	Ease and feasibility of repairing a component or system.	repairability; Repair/maintenance/replaceability	20	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-36	Service, maintenance, and replacement	replaceability	Ease and feasibility of replacing a component without disproportionate disturbance.	replaceability; Repair/maintenance/replaceability	8	non-Core	Requirement known, provenance weak	The attribute is repeatedly required, but implementation/provenance evidence is weak or absent.
ATT-37	Reuse, recycling, and recovery	reuse potential	Potential for a component or material to be reused in the same or another application.	reuse potential	562	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.

ID	Family	Attribute	Information meaning	Original terminology	Frequency n	Readiness	Provenance basis	Notes
ATT-38	Connections and interfaces	reversible connection	Connection that can be undone without unacceptable damage to joined parts.	reversible connection; Connection reversibility/separability	119	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-39	Scenario and expert inputs	scenario assumptions	Explicit assumptions describing future use, lifetime, recovery, market, or system conditions.	scenario assumptions; Scenario/expert assumptions	280	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.
ATT-40	Connections and interfaces	separability	Ability to separate materials or components without contamination or destructive loss.	separability; Connection reversibility/separability	27	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-41	Service, maintenance, and replacement	service life	Expected or actual period over which a building, component, or material remains functional.	service life	273	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-42	Design adaptability and standardization	standardization	Use of standard sizes, interfaces, products, or repeated component rules.	standardization; Modularity/standardization	449	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-43	Supply, take-back, and market conditions	supply-chain information	Supplier availability, sourcing, provenance, logistics, and procurement information.	supply-chain information; Supply chain/transport/take-back	279	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-44	Supply, take-back, and market conditions	take-back information	Supplier, manufacturer, or contractual arrangements for accepting products or materials back.	take-back information; Supply chain/transport/take-back	20	non-Core	Scenario-dependent	The attribute can be operationalized, but its value remains conditional on future conditions, assumptions, or judgment.
ATT-45	Environmental and resource flows	transport	Distance, mode, route, or logistics information for material and product movement.	transport; Supply chain/transport/take-back	234	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.
ATT-46	Material identity and composition	virgin material	Share or quantity of virgin feedstock in a material or product.	virgin material; Circular material content	27	non-Core	Requirement known, provenance weak	At least one implementation was detected, but source and timing reporting are insufficient.
ATT-47	Geometry and spatial definition	volume	Three-dimensional quantity of space, material, or component geometry.	volume; Area and volume	372	Core	Core: retained from v06	Outside the targeted 30-attribute provenance audit; retained without change.
ATT-48	Environmental and resource flows	waste	Expected or actual waste quantity, type, or rate across lifecycle activities.	waste; Waste generation	610	non-Core	Data-source known, stage unresolved	Applied studies report how the input was obtained, but design-stage timing is not established consistently.

Table S4. Focused provenance/readiness audit summary.

Attribute	Required	Implemented	Source reported	Timing evidenced	Prospective	Best P-level	Final status	Outcome	Representative evidence	Note
waste	610	46	23	8	2	P1	non-Core	Data-source known, stage unresolved	R00702 (P1): bim model; R01028 (P1): bim model; secondary data; R00164 (P2): gis	
reuse potential	562	18	10	5	2	P1	non-Core	Scenario-dependent	R00902 (P1): material passport; R01206 (P1): user input; R00235 (P2): material passport	
economic value	557	5	0	0	0	P4	non-Core	Requirement known, provenance weak	R00012 (P4): Not reported; R00016 (P4): building information model; R00030 (P4): Not reported	
LCA Data	377	23	19	7	3	P1	Core	Core: direct stage evidence	R00788 (P1): generic EPDs from One Click LCA; material quantities from a Revit model; R00851 (P1): EPDs and One Click LCA; BIM/Revit quantities and bill of quantities; R00898 (P1): ecoinvent; material specification; bim model	LCA Data moved into Core after the focused audit.
embodied carbon	362	39	19	5	3	P1	non-Core	Data-source known, stage unresolved	R00225 (P1): ice database; price database; building information model; R00788 (P1): environmental product declaration; epds; one click lca; lca database; building information model; R01010 (P1): epd; bill of quantities; bim model	
accessibility	316	11	4	1	0	P2	non-Core	Data-source known, stage unresolved	R00223 (P2): epds; R00249 (P3): floor plan; R00787 (P3): visual inspection	
deconstruction potential	312	10	8	1	1	P1	non-Core	Data-source known, stage unresolved	R00772 (P1): material passport; R00292 (P3): bill of quantity; R00415 (P3): building information model	

Attribute	Required	Implemented	Source reported	Timing evidenced	Prospective	Best P-level	Final status	Outcome	Representative evidence	Note
recycling potential	288	19	14	9	2	P1	non-Core	Scenario-dependent	R00715 (P1): predemolition audit; demolition audit; R00885 (P1): material passport; R00047 (P2): material passport; building material passport	
flexibility	281	11	5	4	1	P1	non-Core	Data-source known, stage unresolved	R00090 (P1): bim model; R00124 (P2): material take-off; bim model; R00164 (P2): gis	
scenario assumptions	280	8	7	2	1	P1	non-Core	Scenario-dependent	R00715 (P1): end-of-life scenario; R00780 (P2): end-of-life scenario; R00420 (P3): future scenario	
supply-chain information	279	2	2	0	0	P3	non-Core	Data-source known, stage unresolved	R00649 (P3): digital product passport; product passport; R01134 (P3): bill of quantities; building information model; bim model; R00129 (P4): Not reported	
service life	273	17	6	2	1	P1	non-Core	Data-source known, stage unresolved	R00851 (P1): environmental product declaration; epds; one click lca; lca database; manufacturer data; R00219 (P2): user input; R00321 (P3): visual inspection	
adaptability	251	11	5	2	0	P2	non-Core	Data-source known, stage unresolved	R00235 (P2): material passport; ifc; R01010 (P2): bim model; R00055 (P3): epds	
expert judgment	250	4	4	0	0	P3	non-Core	Scenario-dependent	R00007 (P3): expert judgment; R00110 (P3): expert opinion; R00223 (P3): epds; expert judgement	
transport	234	12	7	3	2	P1	non-Core	Data-source known, stage unresolved	R00920 (P1): secondary data; R01072 (P1): gis; R00939 (P2): openlca	
disassembly potential	193	11	3	1	1	P1	non-Core	Data-source known, stage unresolved	R00455 (P1): bim model; R00420 (P3): future scenario; R00439 (P3): building information model	

Attribute	Required	Implemented	Source reported	Timing evidenced	Prospective	Best P-level	Final status	Outcome	Representative evidence	Note
manufacturer information	165	3	3	1	0	P2	non-Core	Data-source known, stage unresolved	R00223 (P2): epds; ecoinvent; manufacturer data; R00075 (P3): epd; technical datasheet; manufacturer data; R00520 (P3): manufacturer information	
material toxicity	153	5	3	1	0	P2	non-Core	Data-source known, stage unresolved	R00885 (P2): pre-demolition audit; demolition audit; R00223 (P3): Cradle to Cradle classification; technical sheets; material passports; EPDs; R00981 (P3): bim model	
material passport	122	20	20	5	3	P1	non-Core	Data-source known, stage unresolved	R00772 (P1): material passport; R00885 (P1): material passport; R00902 (P1): material passport	
recovery potential	88	8	3	0	0	P3	non-Core	Scenario-dependent	R00795 (P3): material takeoff; bim model; R00799 (P3): one click lca; R00982 (P3): material passport	
product passport	61	6	6	0	0	P3	non-Core	Data-source known, stage unresolved	R00055 (P3): epds; product passport; R00216 (P3): digital product passport; product passport; R00250 (P3): digital product passport; product passport	
renewable content	53	0	0	0	0	P4	non-Core	Requirement known, provenance weak	R00689 (P4): bim model; R00900 (P4): Not reported; R00922 (P4): Not reported	
recycled content	38	1	0	0	0	P4	non-Core	Requirement known, provenance weak	R00097 (P4): Not reported; R00108 (P4): quantity take-off; building information model; R00139 (P4): Not reported	
separability	27	1	0	0	0	P4	non-Core	Requirement known, provenance weak	R00047 (P4): bim model; R00116 (P4): Not reported; R00212 (P4): Not reported	
virgin material	27	3	0	0	0	P4	non-Core	Requirement known, provenance weak	R00130 (P4): manufacturer data; R00151 (P4): future scenario; R00251 (P4): Not reported	

Attribute	Required	Implemented	Source reported	Timing evidenced	Prospective	Best P-level	Final status	Outcome	Representative evidence	Note
future value	22	0	0	0	0	P4	non-Core	Scenario-dependent	R00373 (P4): Not reported; R00795 (P4): Not reported; R00993 (P4): material passport; building information model	
repairability	20	2	1	0	0	P3	non-Core	Requirement known, provenance weak	R00954 (P3): quantity take-off; bim model; ifc; R00002 (P4): Not reported; R00181 (P4): Not reported	
take-back information	20	0	0	0	0	P4	non-Core	Scenario-dependent	R00279 (P4): Not reported; R00288 (P4): material passport; R00413 (P4): Not reported	
maintainability	19	1	0	0	0	P4	non-Core	Requirement known, provenance weak	R00181 (P4): Not reported; R00240 (P4): Not reported; R00436 (P4): Not reported	
replaceability	8	0	0	0	0	P5	non-Core	Requirement known, provenance weak	R00125 (P5): Not reported; R00139 (P5): Not reported; R00321 (P5): Not reported	

Table S5. AI function, role, evidence-grade, early-design, and Function 6 codebook.

Category	Term	Operational definition	Example/boundary
AI function	Extract/Retrieve	Identify, extract, retrieve, parse, structure, or recognize information already present in a drawing, model, document, passport, database, or specification.	Extract material or EPD fields from a product document.
AI function	Infer/Complete	Infer or fill missing, implicit, incomplete, or unstructured information.	Complete missing IFC properties or infer component relationships.
AI function	Predict/Assess	Produce a predicted or estimated outcome.	Predict embodied carbon or reuse potential.
AI function	Diagnose	Identify drivers, causes, influential variables, limiting factors, or explanatory factors for an outcome.	Explain carbon drivers using SHAP while keeping input variables as diagnostic factors.
AI function	Recommend	Explicitly suggest or select a material, component, strategy, design action, or preferred alternative.	Recommend a product meeting stated environmental criteria.
AI function	Generate plus Rank/Optimize	Umbrella reporting function for generation-only, ranking/optimization-only, and integrated generation-plus-evaluation workflows; the detailed mode is retained separately.	Generate prefabricated alternatives and optimize them against carbon and standardization criteria.
Attribute role	Target/Output	The AI acts on the normalized attribute itself; eligible for A/B evidence.	Target/output status is required for A/B evidence.

Category	Term	Operational definition	Example/boundary
Attribute role	Input/Feature	The attribute is supplied to another AI target; it does not establish target/output capability for the input attribute.	Target/output status is required for A/B evidence.
Attribute role	Criterion	The attribute evaluates or ranks another output.	Target/output status is required for A/B evidence.
Attribute role	Design Variable	The attribute is varied during design search.	Target/output status is required for A/B evidence.
Attribute role	Constraint	The attribute restricts feasible generation or optimization.	Target/output status is required for A/B evidence.
Attribute role	Diagnostic Factor	The attribute helps explain another outcome; it is not itself diagnosed.	Target/output status is required for A/B evidence.
Attribute role	Context Only	The attribute is mentioned but does not perform an operative role.	Target/output status is required for A/B evidence.
Attribute role	Unclear	The full text does not resolve the role; no target/output promotion is permitted.	Target/output status is required for A/B evidence.
Evidence grade	A	At least one verified Direct circularity target/output relationship for the exact attribute-function cell.	Grades describe evidence type, not study quality or deployment maturity.
Evidence grade	B	No A evidence and at least one verified Transferable building-design target/output relationship for the exact cell.	Grades describe evidence type, not study quality or deployment maturity.
Evidence grade	C	No A/B evidence; plausibility is supported only by a verified same-attribute/same-function supporting role.	Grades describe evidence type, not study quality or deployment maturity.
Evidence grade	D	No qualifying A/B evidence and no defensible C basis in the locked corpus; not proof of technical impossibility.	Grades describe evidence type, not study quality or deployment maturity.
Function 6 mode	Generate-only	The implemented workflow generates alternatives but does not verify ranking, optimization, or selection of those alternatives.	Do not infer integrated generation and ranking from optimization alone.
Function 6 mode	Rank/Optimize-only	The implemented workflow ranks, selects, searches, or optimizes existing alternatives or variables without a verified generation step.	Do not infer integrated generation and ranking from optimization alone.
Function 6 mode	Integrated Generate plus Rank/Optimize	The same implemented workflow generates alternatives and evaluates, ranks, optimizes, or selects those generated alternatives.	Do not infer integrated generation and ranking from optimization alone.
Stage marker	Early-design verified	At least one verified target/output relationship for the cell was demonstrated in an explicit or project-status-defensible Conceptual, Schematic, Design Development, or early-inclusive multi-stage context.	The marker describes AI study context and does not change the independent information-readiness classification.
Coding error avoided	Geometry as prediction input	Geometry used to predict embodied carbon remains Input/Feature.	It does not establish Geometry x Predict/Assess.
Coding error avoided	LCA data as criterion	LCA data used to evaluate alternatives remains Criterion.	It does not establish LCA Data x Recommend.
Coding error avoided	Connection type as diagnostic factor	Connection type used to explain reuse remains Diagnostic Factor.	It does not establish Connection Type x Diagnose.

Category	Term	Operational definition	Example/boundary
Coding error avoided	Component count as constraint	Component count restricting generation remains Constraint.	It does not establish Component Count x Generate plus Rank/Optimize.

Table S6. Index of 107 full-text-verified empirical AI papers.

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-00014	Favilli A.; Laccone F.; Cignoni P.; Malomo L.; Giorgi D. (2024). Single-instance, multi-target learning of 3D architectural gridshells for material reuse and circular economy. CEUR Workshop Proceedings.		Direct	Not Reported	Input gridshell; reclaimed-beam stock; Reclaimed stock	Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Full text | PDF p. 4, Full text
AIEXP-00015	Oktar Z.S.; Özener O.Ö. (2026). Urban Circularity Exchange: A BIM-linked Marketplace for Reclaimed Building Materials Powered by AI. Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe. https://doi.org/10.52842/conf.ecaade.2026.3.097	10.52842/conf.ecaade.2026.3.097	Direct	Not Reported	BIM; BIM query; material-passport inventory; Material passport	BIM/IFC computational workflow	Extract/Retrieve; Generate plus Rank/Optimize	PDF p. 6, Full text | PDF p. 3, Full text | PDF p. 1, Full text
AIEXP-00023	Wang T. (2025). Smart mapping of recyclable construction materials using UAV remote sensing and optimised YOLOv8 deep learning within a GIS framework. Microchemical Journal. https://doi.org/10.1016/j.microc.2025.115185	10.1016/j.microc.2025.115185	Direct	Not Reported	Image/photo; UAV image/photo	Machine learning	Extract/Retrieve; Predict/Assess	PDF p. 3, Full text | PDF p. 1, Full text | PDF p. 10, Full text | PDF p. 14, Full text | PDF p. 5, Full text
AIEXP-00034	Cha G.-W.; Park C.-W.; Kim Y.-C. (2024). Optimal Machine Learning Model to Predict Demolition Waste Generation for a Circular Economy. Sustainability (Switzerland). https://doi.org/10.3390/su16167064	10.3390/su16167064	Direct	Not Reported	Building data; Project and building variables	Machine learning	Predict/Assess	PDF p. 1, Full text | PDF p. 2, Full text
AIEXP-00038	Travnicek P.; Nezerka V. (2025). AI-Assisted IoT Monitoring for Tracking Construction and Demolition Waste. 2025 10th International Conference on Smart and Sustainable Technologies, SpliTech 2025. https://doi.org/10.23919/splitech65624.2025.11091388	10.23919/SpliTech65624.2025.11091388	Direct	Not Reported	Sensor/operational data	Implemented computational workflow	Extract/Retrieve	PDF p. 1, Full text
AIEXP-00069	Al-Atawneh Y.; Ismail R.; Albtoush F.; Aljamal M.; Hashemal-Mattarneh; Arrasheed E. (2026). An AI-BIM Based Predictive Framework for the Optimization of Construction Waste Recycling in Saudi Arabia. 2026 International Conference on Artificial Intelligence for Sustainable Engineering and Innovation, AISEI 2026. https://doi.org/10.1109/aisei68628.2026.11572970	10.1109/AISEI68628.2026.11572970	Direct	Not Reported	BIM; project data	Computational optimization; BIM/IFC computational workflow	Predict/Assess	PDF p. 2, Full text
AIEXP-00110	Shen Y.; Wang X.; Lin C.-H.; Li S.; Pei W.; Stouffs R. (2025). Forecasting Future Construction Demands: A Deep Learning approach for material stock circularity analysis of Singapore residential buildings. Proceedings of the International Conference on Computer-Aided Architectural Design Research in Asia. https://doi.org/10.52842/conf.caadria.2025.3.499	10.52842/conf.caadria.2025.3.499	Direct	Not Reported	Building-stock and historical data	Machine learning	Predict/Assess	PDF p. 1, Full text
AIEXP-00126	Trávníček P.; Nežerka V. (2026). LPWAN Smart Waste Bin With On-Device AI Trained on Synthetic Data. IEEE Journal of Radio Frequency Identification. https://doi.org/10.1109/jrfid.2026.3667562	10.1109/JRFID.2026.3667562	Direct	Not Reported	Image/photo	Implemented computational workflow	Extract/Retrieve	PDF p. 1, Full text

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-00139	Tajadod O.E.; Ravanshadnia M.; Ghanbari M. (2025). Integrating circular economy and life cycle assessment strategies in climate-resilient buildings: An artificial intelligence approach to enhance thermal comfort and minimize CO2 emissions in Iran. Energy. https://doi.org/10.1016/j.energy.2025.135064	10.1016/j.energy.2025.135064	Direct	Not Reported	LCA database; Parametric material and envelope variables; Parametric model; Parametric model; LCA database	Computational optimization	Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Full text
AIEXP-00183	Porther N.; Nourbakhshrezaei A.; Jadidi M. (2026). A BIM and LLM Framework for Automated Construction and Demolition Waste Management. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives. https://doi.org/10.5194/isprs-archives-xlix-b4-2026-507-2026	10.5194/isprs-archives-XLIX-B4-2026-507-2026	Direct	Not Reported	BIM; IFC; BIM; IFC; text; IFC	Large language model; BIM/IFC computational workflow	Extract/Retrieve; Predict/Assess	PDF p. 4, Full text | PDF p. 1, Full text | PDF p. 2, Full text
AIEXP-00189	Gordon M.; Batallé A.; De Wolf C.; Sollazzo A.; Dubor A.; Wang T. (2023). Automating building element detection for deconstruction planning and material reuse: A case study. Automation in Construction. https://doi.org/10.1016/j.autcon.2022.104697	10.1016/j.autcon.2022.104697	Direct	Not Reported	Point cloud/scan	Implemented computational workflow	Extract/Retrieve	PDF p. 1, Full text | PDF p. 4, Full text
AIEXP-00216	Garcia R.B.; Silva R.P.; Jorge T.S.; Santos J.; Assunção L.; Oliveira P.; Santos R.; Couceiro M.S.; Ribeiro D. (2026). Enabling Circular Reuse of Sandwich Panels Through UAV Inspection, Deep Learning, and BIM-Based Material Passports. Sustainability (Switzerland). https://doi.org/10.3390/su18052454	10.3390/su18052454	Direct	Not Reported	BIM; material passport; UAV image/photo; BIM	Machine learning; BIM/IFC computational workflow	Extract/Retrieve	PDF p. 1, Full text
AIEXP-00246	Arsirii O.; Ivanov O.; Bieliaiev K.; Cudecka-Purina N. (2025). Intelligent Information Technology for Inventory and Utilization of Construction and Demolition Waste From Damaged Infrastructure Facilities. CEUR Workshop Proceedings.		Direct	Not Reported	Image/photo; inventory data; Image/photo; text/document	Implemented computational workflow	Predict/Assess	PDF p. 3, Full text | PDF p. 2, Full text
AIEXP-00261	Moya-Jiménez R.; Goyes-Balladares A.; Moya-Jiménez G.; Medina-Moncayo A.; Chávez-Ortiz B.; Obando-Navas C.; Arias-Granda S. (2026). Application of Computer Vision and Parametric Design Algorithms for the Reuse of Construction Materials. Buildings. https://doi.org/10.3390/buildings16010184	10.3390/buildings16010184	Direct	Not Reported	Classified fragment inventory; Classified fragments; parametric model; Fragment images; parametric model; Image/photo	Computer vision; Generative design; Computational optimization	Extract/Retrieve; Generate plus Rank/Optimize	PDF p. 3, Full text | PDF p. 5, Full text | PDF p. 1, Full text
AIEXP-00262	Płoszaj-Mazurek M.; Tofiluk A. (2026). A workflow for LLM-assisted IFC data enrichment to support circular reuse of large-panel building systems in Poland. Journal of Building Engineering. https://doi.org/10.1016/j.jobe.2026.115397	10.1016/j.jobe.2026.115397	Direct	Not Reported	Enriched IFC; IFC; condition descriptions; IFC; lookup tables; expert oversight	Large language model; BIM/IFC computational workflow	Infer/Complete; Predict/Assess	PDF p. 1, Full text | PDF p. 10, Full text
AIEXP-00349	Gan W.; Li S.; Li J.; Peng S.; Li R.; Qiu L.; Li B.; He Y. (2025). Enhanced Recognition of Sustainable Wood Building Materials Based on Deep Learning and Augmentation. Sustainability (Switzerland). https://doi.org/10.3390/su17156683	10.3390/su17156683	Transferable	Not Reported	Image/photo	Machine learning	Extract/Retrieve	PDF p. 2, Full text

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-00350	Zhu H.; Hu C.; Zhou C.; Wang Z.; Wang X.; Wu Y. (2026). An explainable machine learning framework for multi-objective carbon reduction targeting material operational seasonal emissions in building retrofits. Scientific Reports. https://doi.org/10.1038/s41598-025-29418-z	10.1038/s41598-025-29418-z	Transferable	Not Reported	Material options; Parametric model; Parametric model; material options; Parametric model; simulation output; Simulation output	Explainable machine learning; Machine learning; Computational optimization	Diagnose; Generate plus Rank/Optimize; Predict/Assess	PDF pp. 9 and 12, Methods and Results | PDF p. 13, SHAP analysis of input variables | PDF p. 14, Optimal performance and parametric analysis | PDF pp. 14–18, Optimal performance and parametric analysis | PDF p. 13, SHAP analysis
AIEXP-00361	Balogun H.; Alaka H.; Demir E.; Egwim C.N.; Wusu G.E.; Yusuf W.; Adegoke M.; Qasim I. (2025). Deconstructability prediction for building using machine learning and ensemble feature selection techniques. Scientific Reports. https://doi.org/10.1038/s41598-025-00790-0	10.1038/s41598-025-00790-0	Direct	Not Reported	Questionnaire/project data; Questionnaire/project variables	Machine learning	Diagnose; Predict/Assess	PDF p. 1, Full text | PDF p. 3, Full text
AIEXP-00406	Cabral S.; Klimenka M.; Bademosi F.; Lau D.; Pender S.; Villaggi L.; Stoddart J.; Donnelly J.; Storey P.; Benjamin D. (2025). A Contactless Multi-Modal Sensing Approach for Material Assessment and Recovery in Building Deconstruction. Sustainability (Switzerland). https://doi.org/10.3390/su17020585	10.3390/su17020585	Direct	Not Reported	Magnetic sensing; Thermal-visible images	Machine learning	Extract/Retrieve; Predict/Assess	PDF p. 3, Full text | PDF p. 15, Full text | PDF p. 4, Full text
AIEXP-00408	Vergés R.; Gaspar K.; Forcada N. (2026). Large-scale analysis of environmental and circularity metrics in construction products through automated data extraction using large language models. Environmental Impact Assessment Review. https://doi.org/10.1016/j.eiar.2025.108243	10.1016/j.eiar.2025.108243	Direct	Not Reported	EPD text	Large language model	Extract/Retrieve	PDF p. 3, Full text | PDF p. 4, Full text
AIEXP-00431	Pinto Almeida C.D.; Jordán Buenaño N. (2026). Lightweight deep learning for sustainable adobe construction: a block-level baseline aligned with circular economy principles and sustainable development goals. Frontiers in Built Environment. https://doi.org/10.3389/fbuil.2026.1883353	10.3389/fbuil.2026.1883353	Direct	Not Reported	Image/photo	Machine learning	Extract/Retrieve	PDF p. 2, Full text
AIEXP-00450	Rajaratnam D.; Stewart R.A.; Liu T.; Vieira A.S.; Amarasinghe I. (2025). A Computer Vision Approach for Estimating Embodied Carbon to Support Circular Economy Decisions. 2025 5th International Conference on Advanced Research in Computing: Converging Horizons: Uniting Disciplines in Computing Research through AI Innovation, ICARC 2025 - Proceedings. https://doi.org/10.1109/icarc64760.2025.10963282	10.1109/ICARC64760.2025.10963282	Direct	Not Reported	Image/photo; Image/photo; specification; reference database; Image/reference data; Specification/reference data	Computer vision	Extract/Retrieve; Predict/Assess	PDF p. 1, Full text | PDF p. 5, Full text
AIEXP-00469	Mi S.; Wu X.; Liu S.; Huang Z.; De Wolf C.; El-Assady M.; Wu P.-Y. (2026). Enhancing building product inventory automation through a hybrid spatial-semantic modeling framework. Developments in the Built Environment. https://doi.org/10.1016/j.dibe.2026.100992	10.1016/j.dibe.2026.100992	Direct	End of Life	Image/photo; Image/photo; text prompt; Pre-demolition inventory sheets	Implemented computational workflow	Extract/Retrieve; Infer/Complete; Predict/Assess	PDF p. 4, Sec. 4, Method | PDF p. 3, Sec. 3.1, Data

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-00471	Kairlapova A.; Forth K.; Carrara A.; Borrmann A. (2025). MATERIAL DETECTION AND CLASSIFICATION IN 2D ARCHITECTURAL DRAWINGS USING COMPUTER VISION FOR MATERIAL PASSPORTS. Proceedings of the European Conference on Computing in Construction. https://doi.org/10.35490/ec3.2025.196	10.35490/EC3.2025.196	Direct	Not Reported	Architectural drawing	Computer vision	Extract/Retrieve	PDF p. 2, Methodology | PDF p. 1, Abstract
AIEXP-00474	Fritzsche L.; Starke R.; Vukorep I. (2026). Addressing the Long Tail of Building Component Reuse through Open-Vocabulary Vision: Image-Based Analysis of Building Components in On-Site Audits. Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe. https://doi.org/10.52842/conf.ecaade.2026.1.481	10.52842/conf.ecaade.2026.1.481	Direct	Not Reported	Image/photo	Implemented computational workflow	Extract/Retrieve	PDF p. 1, Full text
AIEXP-00495	Forth K.; De Wolf C. (2026). Semantic property enrichment for circularity assessments using BIM and labeled property graphs. Journal of Building Engineering. https://doi.org/10.1016/j.jobe.2026.115417	10.1016/j.jobe.2026.115417	Direct	Not Reported	BIM; IFC; property graph	BIM/IFC computational workflow	Infer/Complete; Predict/Assess	PDF p. 2, Full text
AIEXP-00497	Forth K.; Berggold P.; Borrmann A. (2024). Domain-Specific Fine-Tuning of LLM for Material Matching of BIM Elements and Material Passports. Computing In Civil Engineering 2024: Building Information Modeling, Digital Twins, and Simulation and Visualization - Selected Papers from the ASCE International Conference on Computing in Civil Engineering 2024. https://doi.org/10.1061/9780784486122.064	10.1061/9780784486122.064	Direct	Not Reported	IFC; material passport	Large language model; BIM/IFC computational workflow	Extract/Retrieve; Infer/Complete	PDF p. 1, Full text
AIEXP-00498	Lopes A.; Teixeira C.A. (2025). Data-Driven Optimization of Construction and Demolition Waste Management: Pattern Recognition and Anomaly Detection. Sustainability (Switzerland). https://doi.org/10.3390/su17094211	10.3390/su17094211	Direct	Not Reported	Waste-management records	Computational optimization	Predict/Assess	PDF p. 2, Full text
AIEXP-00499	Lauble S.; Geiger A.; Zielke P. (2025). Leveraging Artificial Intelligence for Creating Material Passports from Building Stock Documents: A Workflow for Promoting Circularity in the Built Environment. IOP Conference Series: Earth and Environmental Science. https://doi.org/10.1088/1755-1315/1546/1/012076	10.1088/1755-1315/1546/1/012076	Direct	Operation	Architectural drawing; Architectural drawing; text/document; EPD and external databases; Plans; documents; external databases	BIM/IFC computational workflow	Extract/Retrieve	PDF p. 7, Sec. 5, Discussion based on the case study | PDF p. 6, Sec. 4, Presentation of the workflow
AIEXP-00500	Zhang Y.; Peng J.; Wang Z.; Xi M.; Liu J.; Xu L. (2025). Machine Learning-Assisted Sustainable Mix Design of Waste Glass Powder Concrete with Strength–Cost–CO2 Emissions Trade-Offs. Buildings. https://doi.org/10.3390/buildings15152640	10.3390/buildings15152640	Direct	Not Reported	Concrete-mix variables; Cost data; Reference/experimental data	Evolutionary optimization; Machine learning; Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Full text
AIEXP-00505	Wu P.-Y.; Mjörnell K.; Mangold M.; Sandels C.; Johansson T. (2021). A data-driven approach to assess the risk of encountering hazardous materials in the building stock based on environmental inventories. Sustainability (Switzerland). https://doi.org/10.3390/su13147836	10.3390/su13147836	Transferable	Not Reported	Environmental inventory	Implemented computational workflow	Predict/Assess	PDF p. 3, Full text
AIEXP-00533	Rakhshan K.; Morel J.-C.; Daneshkhah A. (2021). A probabilistic predictive model for assessing the economic reusability of load-bearing building components: Developing a Circular Economy framework. Sustainable Production and Consumption. https://doi.org/10.1016/j.spc.2021.01.031	10.1016/j.spc.2021.01.031	Direct	Not Reported	Component/reusability factors; Reusability factors	Explainable machine learning	Diagnose; Predict/Assess	PDF p. 1, Full text | PDF p. 7, Full text

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-00536	Moussavi S.M.; Svatoš-Ražnjević H.; Körner A.; Tahouni Y.; Menges A.; Knippers J. (2022). Design based on availability: Generative design and robotic fabrication workflow for non-standardized sheet metal with variable properties. International Journal of Space Structures. https://doi.org/10.1177/09560599221081104	10.1177/09560599221081104	Direct	Construction; Detailed Design	Image/photo; Parametric model; Parametric model; image/photo; Reclaimed-material scan/database; parametric model	Generative design; Computational optimization	Extract/Retrieve; Generate plus Rank/Optimize	PDF p. 7, Computational workflow | PDF p. 10, Robotic fabrication workflow | PDF p. 6, Computational workflow | PDF p. 8, Search-engine grouping | PDF p. 8, Computational workflow | PDF p. 5, Computational workflow
AIEXP-00565	Płoszaj-Mazurek M.; Ryńska E.; Grochulska-Salak M. (2020). Methods to optimize carbon footprint of buildings in regenerative architectural design with the use of machine learning, convolutional neural network, and parametric design. Energies. https://doi.org/10.3390/en13205289	10.3390/en13205289	Transferable	Conceptual	Parametric design features; Parametric design variables	Machine learning	Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Full text
AIEXP-00637	Hany Y.; Ahmed W.; Elshazly A.; Senousi A.M.; Darwish W. (2025). Automated 3D Reconstruction of Interior Structures from Unstructured Point Clouds. ISPRS International Journal of Geo-Information. https://doi.org/10.3390/ijgi14110428	10.3390/ijgi14110428	Transferable	Operation	Point cloud / scan	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF p. 1, Abstract
AIEXP-00641	Lu Z.; Hu Y. (2026). BIM-based ontology-driven knowledge graph framework for policy-aligned building salvage assessment. Automation in Construction. https://doi.org/10.1016/j.autcon.2026.106985	10.1016/j.autcon.2026.106985	Direct	End of Life	As-built BIM/IFC; BIM; IFC; text/document; Policy/facility documents; Text/document	Large language model; Knowledge graph; BIM/IFC computational workflow	Extract/Retrieve; Infer/Complete; Recommend	PDF p. 11, Sec. 4.1, Data collection and extraction | PDF p. 16, Sec. 5.4 and Table 5 | PDF p. 14, Sec. 5.2, KG integrity results
AIEXP-00800	Li K.; Gan V.J.L.; Li M. (2025). Automating 3D As-Built BIM with Deep Learning and Knowledge-Driven Graphs. Computing in Civil Engineering 2025: Computational and Intelligent Technologies - Selected Papers from the ASCE International Conference on Computing in Civil Engineering 2025. https://doi.org/10.1061/9780784486436.069	10.1061/9780784486436.069	Transferable	Operation	Point cloud / scan; Point cloud / scan; knowledge graph	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF p. 1, Abstract
AIEXP-00847	Won Ma J.; Jung J.; Leite F. (2024). Deep Learning-Based Scan-To-BIM Automation and Object Scope Expansion Using a Low-Cost 3D Scan Data. Journal of Computing in Civil Engineering. https://doi.org/10.1061/jccee5.cpeng-5751	10.1061/JCCEE5.CPENG-5751	Transferable	Operation	Point cloud / scan	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF p. 1, Abstract
AIEXP-00879	Moshari A.; Javanroodi K.; Nik V.M. (2026). LLM-enabled bidirectional material semantic alignment for automated BIM-to-building performance assessment. Automation in Construction. https://doi.org/10.1016/j.autcon.2026.107258	10.1016/j.autcon.2026.107258	Transferable	Conceptual	Cost database; IFC; material database; IFC; material database; LCA/cost criteria; LCA database	Large language model; BIM/IFC computational workflow	Generate plus Rank/Optimize; Infer/Complete	PDF p. 1, Abstract | PDF pp. 1 and 13, Abstract; limitations
AIEXP-00946	Bayer H.; Schönfelder P.; König M. (2026). Deep learning-based structural element extraction and knowledge-graph-based validation of 2D bridge drawings. Automation in Construction. https://doi.org/10.1016/j.autcon.2026.107213	10.1016/j.autcon.2026.107213	Transferable	Operation	Architectural/construction drawing; Drawing; knowledge graph	Optical character recognition; Machine learning	Extract/Retrieve; Infer/Complete	PDF p. 1, Abstract

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-01043	Rahman M.; Adebayo S.; Acevedo-Mejia D.A.; Hester D.; McPolin D.; Rafferty K.; Laefer D.F. (2026). ISC-Perception: A Hybrid Vision Dataset for Robotic Assembly with Novel Intermeshed Steel Connections. Buildings. https://doi.org/10.3390/buildings16173407	10.3390/buildings16173407	Transferable	Construction	Construction images; Image/photo	Implemented computational workflow	Extract/Retrieve	PDF pp. 8 and 16–22, Methods and Results | PDF pp. 8 and 22, Methods and Conclusions
AIEXP-01249	Austern G.; Bloch T.; Abulafia Y. (2024). Incorporating Context into BIM-Derived Data—Leveraging Graph Neural Networks for Building Element Classification. Buildings. https://doi.org/10.3390/buildings14020527	10.3390/buildings14020527	Transferable	Not Reported	BIM; BIM geometry; BIM proximity graph	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF pp. 1 and 15, Abstract and Results | PDF pp. 1–2, Abstract and Introduction | PDF pp. 1 and 13, Abstract and Methods
AIEXP-01354	Kim H.; Kim C. (2021). 3D as-built modeling from incomplete point clouds using connectivity relations. Automation in Construction. https://doi.org/10.1016/j.autcon.2021.103855	10.1016/j.autcon.2021.103855	Transferable	Operation	Point cloud / scan	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF p. 1, Abstract
AIEXP-01509	Karmakar A.; Delhi V.S.K. (2023). Restructuring Elements in IFC Topology through Semantic Enrichment: A Case of Automated Compliance Checking. Proceedings of the International Symposium on Automation and Robotics in Construction. https://doi.org/10.22260/isarc2023/0062	10.22260/ISARC2023/0062	Transferable	Not Reported	IFC; IFC geometry	BIM/IFC computational workflow	Infer/Complete	PDF p. 6, Sec. 5
AIEXP-01537	Kim S.; Yajima Y.; Park J.; Chen J.; Cho Y.K. (2022). A Hybrid Semantic-Geometric Approach for Clutter-Resistant Floorplan Generation from Building Point Clouds. International Conference on Construction Engineering and Project Management.		Transferable	Operation	Point cloud / scan; Point cloud / scan; drawing	Implemented computational workflow	Infer/Complete	PDF pp. 3–4, Secs. 2–3 | PDF p. 3, Sec. 2
AIEXP-01543	Xiong Z.; Wang T. (2021). Research on BIM Reconstruction Method Using Semantic Segmentation Point Cloud Data Based on PointNet. IOP Conference Series: Earth and Environmental Science. https://doi.org/10.1088/1755-1315/719/2/022042	10.1088/1755-1315/719/2/022042	Transferable	Operation	Point cloud / scan	Machine learning; BIM/IFC computational workflow	Infer/Complete	PDF p. 1, Abstract
AIEXP-01636	Stellmacher D.; Lögten K.G.; Theißen S.; Kelm A.; Meins-Becker H.A. (2025). DIGITAL DECISION SUPPORT IN CONSTRUCTION: ANALYSIS OF INFORMATION REQUIREMENTS AND DATA PROVISION FOR AI-BASED SELECTION OF SUSTAINABLE BUILDING PRODUCTS. Proceedings of the European Conference on Computing in Construction. https://doi.org/10.35490/ec3.2025.236	10.35490/EC3.2025.236	Direct	Not Reported	EPD; EPD; product datasheet; EPD; text/document; Knowledge graph; EPD; product datasheet; Manufacturer documentation; product datasheet; Product datasheet; EPD; Product datasheets; manufacturer websites; Safety Data Sheet; product datasheet	Graph retrieval-augmented generation; Large language model	Extract/Retrieve; Infer/Complete; Recommend	PDF pp. 5-7, Information acquisition, processing, and results | PDF pp. 3-6, Tables 1-2; extraction workflow | PDF pp. 3-6, Table 1; extraction workflow | PDF p. 7, Results
AIEXP-01656	Stellmacher D.; Kelm A.; Pomp A.; Meins-Becker A. (2025). AN INNOVATIVE APPROACH TO THE SEMANTIC INTEGRATION OF CONSTRUCTION AND SUSTAINABILITY INFORMATION USING GRAPHRAG. Proceedings of International Structural Engineering and Construction. https://doi.org/10.14455/isec.2025.12(1).con-25	10.14455/ISEC.2025.12(1).CON-25	Direct	Not Reported	EPD; product datasheet; certificate; EPD; text/document; Manufacturer documentation; product datasheet; Product datasheets; manufacturer websites; Safety Data Sheet; product datasheet	Graph retrieval-augmented generation; Large language model	Extract/Retrieve; Infer/Complete	PDF pp. 1-6, Method and results | PDF p. 6, Discussion

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-01687	Abdulrahman A.A.; Kay R. (2024). CAN MACHINE LEARNING AUTOMATE CARBON CLASSIFICATION OF MATERIALS WITHIN A BUILDING INFORMATION MODEL?. Proceedings of the European Conference on Computing in Construction. https://doi.org/10.35490/ec3.2024.229	10.35490/EC3.2024.229	Transferable	Not Reported	BIM; carbon database; BIM; material/carbon database	Machine learning; BIM/IFC computational workflow	Extract/Retrieve; Infer/Complete	PDF p. 1, Abstract
AIEXP-01711	Neupane B.; Hilloulin B.; Chenouard R.; Roziere E. (2026). Quantifying the impact of data quality on machine learning prediction: a large-scale study of construction product environmental declarations. International Journal of Life Cycle Assessment. https://doi.org/10.1007/s11367-026-02676-3	10.1007/s11367-026-02676-3	Direct	Not Reported	EPD dataset	Explainable machine learning; Machine learning	Diagnose; Predict/Assess	PDF p. 1, Abstract | PDF p. 18, Conclusions
AIEXP-01718	Adu-Duodu K.; Wilson S.; Li Y.; Rana O.; Wang Y.; Ranjan R.; Shah T.; Solaiman E. (2026). Human-in-the-loop semantic middleware for construction compliance checking and safe product reuse. Advanced Engineering Informatics. https://doi.org/10.1016/j.aei.2026.105112	10.1016/j.aei.2026.105112	Direct	End of Life	Expert review; Product datasheets; EPD; regulation text; knowledge graph; Product documents; regulation ontology; Study data source	Implemented computational workflow	Diagnose; Generate plus Rank/Optimize; Infer/Complete; Predict/Assess	PDF pp. 12-13, Case Study II | PDF pp. 8 and 13, Human verification; Case Study II | PDF pp. 8 and 15, Human verification layer | PDF p. 1, Abstract
AIEXP-01724	Arslan M.; Munawar S. (2026). Environmental impact analysis of structural steels using retrieval-augmented generation and large language models. Green Technologies and Sustainability. https://doi.org/10.1016/j.grets.2026.100352	10.1016/j.grets.2026.100352	Direct	Conceptual	EPD	Large language model	Extract/Retrieve	PDF pp. 1-2 and 9, Abstract; Sec. 1; Table 9 | PDF p. 2, Sec. 1 | PDF p. 9, Table 9
AIEXP-01731	Billah W.; Abdou A.A.; Budaiwi I.M. (2026). A data-driven framework for environmental impact assessment of building materials using environmental product declarations. Current Research in Environmental Sustainability. https://doi.org/10.1016/j.crsust.2026.100376	10.1016/j.crsust.2026.100376	Direct	Conceptual	333 EPD records; EPD; EPD dataset	Implemented computational workflow	Diagnose; Extract/Retrieve; Predict/Assess	PDF p. 1, Abstract | PDF p. 12, Secs. 3.8-3.9 | PDF p. 3, Sec. 2.1
AIEXP-01743	Stellmacher D.; Lögten K.G.; Theißen S.; Kelm A.; Meins-Becker A. (2025). GRAPHRAG-BASED DECISION SUPPORT SYSTEM INTEGRATING AND SELECTING SUSTAINABLE BUILDING PRODUCTS. Proceedings of International Structural Engineering and Construction. https://doi.org/10.14455/isec.2025.12(1).con-08	10.14455/ISEC.2025.12(1).CON-08	Direct	Not Reported	EPD; text/document; Knowledge graph; EPD; datasheet; Manufacturer documentation; product datasheet; Product datasheet; Product datasheets; manufacturer websites; Safety Data Sheet; product datasheet	Graph retrieval-augmented generation; Large language model	Extract/Retrieve; Infer/Complete; Recommend	PDF pp. 2-6, Secs. 3-5 | PDF pp. 3-4, Method and Table 1 | PDF p. 5, Sec. 4.2
AIEXP-01750	Forth K.; Abualdenien J.; Borrmann A. (2023). Calculation of embodied GHG emissions in early building design stages using BIM and NLP-based semantic model healing. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2023.112837	10.1016/j.enbuild.2023.112837	Transferable	Conceptual	BIM/IFC and LCA knowledge database; BIM; IFC; LCA database	Natural language processing; BIM/IFC computational workflow	Infer/Complete; Predict/Assess	PDF p. 2, Sec. 1, Introduction | PDF p. 1, Abstract | PDF p. 2, Method and nomenclature | PDF p. 15, Method and nomenclature | PDF p. 1, Method and nomenclature
AIEXP-01769	Emunds C.; Frisch J.; van Treeck C. (2025). Spatial Link Prediction: Learning topological relationships in MEP systems. Advanced Engineering Informatics. https://doi.org/10.1016/j.aei.2025.103414	10.1016/j.aei.2025.103414	Transferable	Not Reported	BIM; IFC; graph	Implemented computational workflow	Infer/Complete	PDF pp. 1-2, Abstract; contributions

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-01773	Allam A.S.; Nik-Bakht M. (2026). IFC-based automated deconstructability analysis with market data integration through knowledge graphs. Automation in Construction. https://doi.org/10.1016/j.autcon.2026.107219	10.1016/j.autcon.2026.107219	Direct	End of Life	BIM/IFC and market webpages; BIM; IFC; BIM; IFC; knowledge graph; BIM; IFC; text/document; Text/document; product database	Large language model; Knowledge graph; BIM/IFC computational workflow	Extract/Retrieve; Infer/Complete; Recommend	PDF p. 14, Sec. 4.1, Case study | PDF p. 13, Sec. 4, Framework implementation and evaluation | PDF p. 12, Sec. 3.2, AI-powered semantic enhancement | PDF p. 12, Sec. 3.2, Market-to-building integration
AIEXP-01779	Luo L.; Chen Y.; Feng W.; Wu J.; Cai W. (2025). Progressive prediction of embodied carbon emissions across stages of schematic design with machine learning. Journal of Cleaner Production. https://doi.org/10.1016/j.jclepro.2025.145799	10.1016/j.jclepro.2025.145799	Transferable	Schematic	Design-stage data; Design-stage variables	Machine learning	Predict/Assess	PDF p. 1, Full text
AIEXP-01780	Mao L.; Xiang C. (2026). A cross-climate multi-stage approach to life-cycle carbon sensitivity analysis in BIPV-integrated building design. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2026.117011	10.1016/j.enbuild.2026.117011	Transferable	Multiple	Parametric model; BPS-LCA simulation; Parametric model; simulation output; Study data source	Machine learning; Computational optimization	Diagnose; Generate plus Rank/Optimize; Infer/Complete; Predict/Assess	PDF p. 1, Abstract
AIEXP-01785	Wang Z.; Zhang L.; Pu M. (2026). Project-Level Embodied Carbon Prediction Across Building Design Stages Using a Machine Learning Framework. Sustainability (Switzerland). https://doi.org/10.3390/su18157723	10.3390/su18157723	Transferable	Construction Documentation; Not Reported; Schematic	Project geometric/material variables; Project variables	Explainable machine learning; Machine learning	Diagnose; Predict/Assess	PDF p. 1, Full text
AIEXP-01788	Zhang X.; Chen H.; Sun J.; Zhang X. (2024). Predictive models of embodied carbon emissions in building design phases: Machine learning approaches based on residential buildings in China. Building and Environment. https://doi.org/10.1016/j.buildenv.2024.111595	10.1016/j.buildenv.2024.111595	Transferable	Multiple	Project records; Project records; material quantities	Machine learning	Predict/Assess	PDF p. 13, Sec. 6 Conclusion | PDF pp. 4 and 13, Secs. 3.3 and 6
AIEXP-01789	Al-Habaibeh A.; Manu E.; Shakmak B.; Selvam J.; Lin T.-H.; Clement T. (2024). Rapid Evaluation of Cost and Whole Life Carbon of Buildings Using Artificial Intelligence. Proceedings - 2024 IEEE/ACM International Conference on Big Data Computing, Applications and Technologies, BDCAT 2024. https://doi.org/10.1109/bdcat63179.2024.00037	10.1109/BDCAT63179.2024.00037	Transferable	Not Reported	Building design variables	Implemented computational workflow	Predict/Assess	PDF p. 1, Full text
AIEXP-01796	Wadhwa J.; Bhadouria S.S.; Kumari P. (2025). AI-Driven Multi-Criteria Material Selection and Safety Analysis for Sustainable Building Design in India. 2025 International Conference on Emerging Technologies and Innovation for Sustainability, EmergIN 2025. https://doi.org/10.1109/emergin67762.2025.11450792	10.1109/EmergIN67762.2025.11450792	Transferable	Not Reported	IFC India database; Material database; simulation output; Material database; site/design scenarios	Implemented computational workflow	Generate plus Rank/Optimize; Recommend	PDF p. 1, Abstract
AIEXP-01811	Guo D.; Di Modica P.; La Rosa A.D. (2025). A Semantic-enriched LCI database for the Embodied Environmental Evaluation of Buildings through Knowledge Graph Technologies. IFAC-PapersOnLine. https://doi.org/10.1016/j.ifacol.2025.09.416	10.1016/j.ifacol.2025.09.416	Direct	Multiple	LCI database; knowledge graph; LCI/LCA database; knowledge graph; Study data source	Knowledge graph	Infer/Complete	PDF p. 1, Abstract
AIEXP-01821	Iqbal N.; Shabbir K.; Noureldin M. (2025). A novel explainable AI-based design optimization framework to estimate sustainability and economic impacts of reinforced concrete structures. Machine Learning with Applications. https://doi.org/10.1016/j.mlwa.2025.100760	10.1016/j.mlwa.2025.100760	Transferable	Conceptual	ML/XAI counterfactual workflow; Parametric dataset; project data	Explainable machine learning; Machine learning; Computational optimization	Diagnose; Predict/Assess; Recommend	PDF p. 1, Abstract | PDF pp. 8 and 24, Secs. 2.4.3 and 5 | PDF p. 24, Sec. 5 Conclusions | PDF pp. 23-24, Counterfactual results; Sec. 5

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-01838	Pournaghshband A.; Piadeh F.; Ahmadi M.; Sahaf A. (2026). Modular multi-domain AI framework for sustainable construction material optimisation. Environmental Impact Assessment Review. https://doi.org/10.1016/j.eiar.2026.108582	10.1016/j.eiar.2026.108582	Direct	Conceptual	Cost-related LCA outputs; EoL scenarios; LCA database; LCA database; product/material scenarios; LCA database; structured scenario table	Computational optimization	Generate plus Rank/Optimize; Predict/Assess; Recommend	PDF pp. 1 and 9, Abstract and Methods | PDF pp. 9 and 15, Optimization and Results | PDF p. 15, Results and Discussion | PDF p. 15, Results | PDF pp. 7 and 15, Methods and Results
AIEXP-01859	Vlasenko T.; Hutsol T.; Vlasovets V.; Glowacki S.; Nurek T.; Horetska I.; Kukharets S.; Firman Y.; Bilovod O. (2025). Ensemble learning based sustainable approach to rebuilding metal structures prediction. Scientific Reports. https://doi.org/10.1038/s41598-024-84996-8	10.1038/s41598-024-84996-8	Direct	Not Reported	Non-destructive magnetic data	Implemented computational workflow	Predict/Assess	PDF p. 1, Full text
AIEXP-01864	Wang B.; Lu W.; Li X.; Jin F. (2026). A graph neural network-enhanced surrogate model for rapid optimization of carbon-efficient building thermal insulation systems. Journal of Building Engineering. https://doi.org/10.1016/j.jobe.2026.115234	10.1016/j.jobe.2026.115234	Transferable	Not Reported	Graph representation; Parametric model; GNN surrogate; Parametric model; simulation output	Evolutionary optimization; Machine learning; Computational optimization	Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Abstract
AIEXP-01871	Budig M.; Heckmann O.; Boon A.N.Q.; Hudert M.; Lork C.; Cheah L. (2020). Data-driven Embodied Carbon Evaluation of Early Building Design Iterations. RE: Anthropocene, Design in the Age of Humans - Proceedings of the 25th International Conference on Computer-Aided Architectural Design Research in Asia, CAADRIA 2020.		Transferable	Conceptual	Floor plan; Floor plan; parametric model; Floor plan; reference data	Machine learning	Predict/Assess	PDF p. 1, Full text
AIEXP-01872	Zawaideh F. (2026). BIM-integrated explainable machine learning for cost, schedule, and carbon risk prediction in high-rise building construction. Asian Journal of Civil Engineering. https://doi.org/10.1007/s42107-026-01761-0	10.1007/s42107-026-01761-0	Transferable	Not Reported	BIM; IFC; BIM; IFC; parametric scenarios	Explainable machine learning; Evolutionary optimization; Machine learning	Diagnose; Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Abstract | PDF pp. 12-14, Results and sensitivity analysis | PDF pp. 13-14, Multi-objective optimization
AIEXP-01875	Torabi M.; Norouzi M.; Evins R. (2026). A machine learning-based surrogate model to approximate building carbon footprint in early-stage architectural design. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2026.117362	10.1016/j.enbuild.2026.117362	Transferable	Conceptual	Parametric LCA; simulation output; Parametric model	Machine learning	Predict/Assess	PDF p. 1, Full text
AIEXP-01888	Gai M.; Kam K.J.; Flor J.-F.; Chai C.; Gunasagaran S. (2026). A Generative AI Framework for Carbon-Oriented Biomimetic Façade Design in Architecture. Buildings. https://doi.org/10.3390/buildings16112082	10.3390/buildings16112082	Transferable	Conceptual	Material carbon factors; simulation output; Material options; Parametric model; simulation output	Generative design	Generate plus Rank/Optimize; Predict/Assess	PDF p. 8, Sec. 2.2 | PDF pp. 14–15, Secs. 2.8–2.9 | PDF pp. 13–17, Secs. 2.7 and 3.2 | PDF pp. 16–17, Secs. 3 and 3.2 | PDF pp. 15–17, Secs. 2.9–3.2
AIEXP-01902	Hsiang S.-H.; Shih S.-G. (2025). Application of Hybrid AI Technology in Calculating Carbon Emissions in Interior Design: Optimizing designers' efficiency and accuracy in sustainable design decision-making. Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe.		Transferable	Not Reported	Carbon-emission database; LCA database; material database; Material and carbon database; Material database; LCA database	Implemented computational workflow	Predict/Assess; Recommend	PDF p. 5, Results | PDF p. 9, Discussion and conclusion

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-01907	Li M.; Pang H.; Yao S.; Yuan J.; Feng Y.; Jiao F. (2026). Cascaded generative design framework for modular prefabricated buildings involving multi-performance response. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2026.117818	10.1016/j.enbuild.2026.117818	Transferable	Conceptual	Parametric model; Parametric model and simulations; Parametric model; simulation output	Evolutionary optimization; Generative design; Computational optimization	Generate plus Rank/Optimize	PDF p. 5, Sec. 2.1.4, Generative framework | PDF p. 6, Sec. 2.2, Multi-performance decision-making | PDF p. 6, Sec. 2.1.4, Generative framework
AIEXP-02074	Bhagat R.M.; Pande P.B.; Madurwar K.V.; Raut J.M.; Vairagade V.S. (2025). Life cycle assessment of 3D-printed building materials towards sustainability-driven optimization and environmental impact analysis using computational intelligence techniques. International Journal of Life Cycle Assessment. https://doi.org/10.1007/s11367-025-02530-y	10.1007/s11367-025-02530-y	Transferable	Not Reported	Material and LCA data; Material database; LCA database	Explainable machine learning; Computational optimization	Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Abstract | PDF p. 1, Abstract and framework summary
AIEXP-02103	Meng X.; Chang J.; Liu X.; Zhuang F. (2026). Classifying the Reuse Value of Industrial Heritage Sites Using Random Forest: A Case Study of Jiangsu's Salt Reclamation Zone. Buildings. https://doi.org/10.3390/buildings16040796	10.3390/buildings16040796	Direct	Not Reported	Expert and heritage-site data	Machine learning	Predict/Assess	PDF p. 1, Full text
AIEXP-02216	Ye F.; Demartino C.; Contiguglia C.P.; Albanesi T.; Wang X.; Xiao Y. (2026). Optimizing bio-based SCM concrete for sustainable construction through structural and environmental evaluation driven by explainable surrogate modeling. Construction and Building Materials. https://doi.org/10.1016/j.conbuildmat.2026.147552	10.1016/j.conbuildmat.2026.147552	Direct	Not Reported	Literature database; LCA factors; Literature database; carbon factors	Explainable machine learning; Evolutionary optimization	Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-02243	Čantrak M.; Micić D.; Prokić D.; Pezo L.; Ćurčić L.; Mladenović M. (2024). Construction Waste Modeling for the Republic of Serbia. Journal of Scientific and Industrial Research. https://doi.org/10.56042/jsir.v83i5.2474	10.56042/jsir.v83i5.2474	Direct	Not Reported	Historical waste data	Implemented computational workflow	Predict/Assess	PDF p. 1, Full text
AIEXP-02540	Zhang W.; Husseini B.; Smerat A.; Elkotb M.A. (2026). Robust frameworks for assessing volumetric carbon footprint of ground granulated blast furnace slag geopolymer concretes. Materials Today Advances. https://doi.org/10.1016/j.mtadv.2026.100923	10.1016/j.mtadv.2026.100923	Direct	Not Reported	Literature-derived mixture database; LCA factors	Explainable machine learning	Diagnose; Predict/Assess	PDF p. 1, Abstract
AIEXP-02655	Wang H.; Liu S.; Luo Z.; Zhou X.; Ren J.; Qian Y. (2026). Assessing carbon emissions from prefabricated component production: Utilizing a Chinese factory dataset for explainable AI-driven prediction. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2026.117305	10.1016/j.enbuild.2026.117305	Transferable	Conceptual	Factory dataset	Explainable machine learning; Machine learning	Diagnose; Predict/Assess	PDF p. 1, Abstract | PDF pp. 9-12, Sec. 3.3.2
AIEXP-02680	Huang X.; Huang J.; Kaewunruen S. (2025). An explainable machine learning system for efficient use of waste glasses in durable concrete to maximise carbon credits towards net zero emissions. Waste Management. https://doi.org/10.1016/j.wasman.2024.12.034	10.1016/j.wasman.2024.12.034	Direct	Not Reported	Experimental dataset; Experimental waste-glass concrete dataset	Explainable machine learning; Machine learning	Diagnose; Predict/Assess	PDF p. 1, Abstract
AIEXP-02802	Kiran K., V (2026). A physics-informed probabilistic machine learning framework for mix design optimisation of waste-based cementitious composites: Uncertainty quantification, SHAP interpretability, and multi-objective pareto analysis. Next Materials. https://doi.org/10.1016/j.nxmate.2026.103214	10.1016/j.nxmate.2026.103214	Direct	Not Reported	Experimental dataset; Material database; experimental data	Explainable machine learning; Machine learning; Computational optimization	Generate plus Rank/Optimize; Predict/Assess	PDF p. 1, Abstract

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-03005	Reisinger J.; Kugler S.; Kovacic I.; Knoll M. (2022). Parametric Optimization and Decision Support Model Framework for Life Cycle Cost Analysis and Life Cycle Assessment of Flexible Industrial Building Structures Integrating Production Planning. Buildings. https://doi.org/10.3390/buildings12020162	10.3390/buildings12020162	Transferable	Not Reported	Cost data; LCA database; Parametric model; Parametric model; LCA/LCC data	Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Full text | PDF p. 2, Full text
AIEXP-03048	El Amaireh A.M.; AlZoubi R.S. (2026). A metaheuristic-driven framework for sustainable material selection in energy-conscious infrastructure projects. Asian Journal of Civil Engineering. https://doi.org/10.1007/s42107-025-01495-5	10.1007/s42107-025-01495-5	Transferable	Not Reported	Material database; expert data; Material database; expert-weighted objectives	Computational optimization	Generate plus Rank/Optimize; Recommend	PDF p. 12, Results and decision maps
AIEXP-03140	Wu H.; Yu X.; Liu W.; Liu S.; Oteng D.; Bao Z. (2026). An automated BIM-driven framework for early-stage prediction and influencing factor analysis of construction and demolition waste from masonry infill walls. Case Studies in Construction Materials. https://doi.org/10.1016/j.cscm.2026.e06173	10.1016/j.cscm.2026.e06173	Direct	Design Development	Revit BIM; Revit BIM; material parameters	BIM/IFC computational workflow	Diagnose; Predict/Assess	PDF p. 1, Full text | PDF p. 11, Full text
AIEXP-03149	Zhang X.; Zhang X. (2024). Low-carbon design optimization of reinforced concrete building structures using genetic algorithm. Journal of Asian Architecture and Building Engineering. https://doi.org/10.1080/13467581.2023.2278466	10.1080/13467581.2023.2278466	Transferable	Not Reported	Structural model; carbon inventory; Structural model; material database; Study data source	Evolutionary optimization; Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-03166	Xiang Y.; Mahamadu A.-M.; Florez-Perez L.; Wu Y. (2024). Design optimisation towards lower embodied carbon of prefabricated buildings: Balancing standardisation and customisation. Developments in the Built Environment. https://doi.org/10.1016/j.dibe.2024.100413	10.1016/j.dibe.2024.100413	Direct	Design Development	Carbon calculation; Parametric model; Parametric model; BIM; Parametric model; genetic algorithm	Evolutionary optimization; Computational optimization	Generate plus Rank/Optimize	PDF pp. 1 and 6, Abstract; Sec. 3.1.2 | PDF p. 1, Abstract | PDF pp. 1 and 4, Abstract; Sec. 3 Methodology | PDF pp. 1 and 18, Abstract; conclusions | PDF pp. 6 and 15, Sec. 3.1.2; results
AIEXP-03172	Pei W.; Li Y.; Ang Y.Q.; Stouffs R. (2026). Design with Adaptability for Circular Building: A Graph-Informed Generative Framework. Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe. https://doi.org/10.52842/conf.ecaade.2026.3.011	10.52842/conf.ecaade.2026.3.011	Direct	Conceptual	Floor plan; graph model; Graph metrics; Graph model; User-defined future living scenarios	Generative design	Generate plus Rank/Optimize	PDF p. 1, Abstract | PDF p. 9, Results | PDF pp. 1 and 9, Abstract and Results | PDF p. 6, Future resilience metric
AIEXP-03174	Sanchez B.; Rausch C.; Haas C.; Saari R. (2020). A selective disassembly multi-objective optimization approach for adaptive reuse of building components. Resources, Conservation and Recycling. https://doi.org/10.1016/j.resconrec.2019.104605	10.1016/j.resconrec.2019.104605	Direct	End of Life	BIM, LCA, and cost data; BIM; LCA database; cost data	Computational optimization	Generate plus Rank/Optimize	PDF p. 3, Sec. 3, Methodology
AIEXP-03195	Sapitri D.; Chang Y.-T.; Hsieh S.-H. (2026). Evaluating Carbon Payback Period in Parametric Facade Optimization Using Genetic Algorithms and Multi-Criteria Decision-Making. Proceedings of the International Symposium on Automation and Robotics in Construction. https://doi.org/10.22260/isarc2026/0240	10.22260/ISARC2026/0240	Transferable	Conceptual	LCA data; LCA factors; Parametric model; LCA factors; Parametric model; LCA factors; simulation output; Parametric model; simulation output	Evolutionary optimization; Computational optimization	Generate plus Rank/Optimize; Recommend	PDF pp. 4–5, Secs. 3.2–3.4 | PDF p. 5, Sec. 3.4
AIEXP-03196	Wang X.; Hu M.; Wen S. (2025). Research on Algorithms for Digital Design and Optimisation of Prefabricated Building Structures. 2025 IEEE International Conference on Electronics, Energy Systems and Power Engineering, EESPE 2025. https://doi.org/10.1109/eespe63401.2025.10986992	10.1109/EESPE63401.2025.10986992	Transferable	Not Reported	BIM; genetic algorithm; BIM; optimization model; Study data source	Evolutionary optimization; Computational optimization; BIM/IFC computational workflow	Generate plus Rank/Optimize; Infer/Complete	PDF pp. 5-6, Experiment and case analysis | PDF p. 6, Experiment and case analysis | PDF p. 1, Abstract

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-03198	Cudzik J.; Nessel M. (2025). Algorithmic space optimization in building adaptations: Generative design comparison. Journal of Building Engineering. https://doi.org/10.1016/j.jobe.2025.113660	10.1016/j.jobe.2025.113660	Transferable	Conceptual	Existing-building floor boundaries; Generated layouts; Parametric model	Generative design; Computational optimization	Generate plus Rank/Optimize	PDF pp. 1 and 5, Abstract; method | PDF p. 1, Abstract
AIEXP-03203	Sajjadian S.M. (2025). Carbon-optimised refurbishment measures for a simulated house in Edinburgh. Intelligent Buildings International. https://doi.org/10.1080/17508975.2025.2480114	10.1080/17508975.2025.2480114	Transferable	Not Reported	Product/material data; simulation; Simulation output; material data; Study data source	Implemented computational workflow	Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-03217	Abrishami S.; Ramos Boada M. (2026). BIM-Integrated Multi-Objective Optimisation of Prefabricated Construction Configurations: A WBS-Based Generative Decomposition Framework. Buildings. https://doi.org/10.3390/buildings16122373	10.3390/buildings16122373	Transferable	Conceptual	BIM; component database; ICE database; BIM	Evolutionary optimization; Generative design; Computational optimization	Generate plus Rank/Optimize	PDF p. 3, Sec. 2.2 | PDF pp. 13 and 17, Secs. 5 and 6 | PDF p. 13, Sec. 5 | PDF pp. 7 and 13, Database and Results
AIEXP-03218	Atashbar H.; Noorzai E. (2023). Optimization of Exterior Wall Cladding Materials for Residential Buildings Using the Non-Dominated Sorting Genetic Algorithm II (NSGAII) Based on the Integration of Building Information Modeling (BIM) and Life Cycle Assessment (LCA) for Energy Consumption: A Case Study. Sustainability (Switzerland). https://doi.org/10.3390/su152115647	10.3390/su152115647	Transferable	Not Reported	BIM; LCA database; BIM; LCA database; simulation output; LCA database	Evolutionary optimization; Computational optimization; BIM/IFC computational workflow	Generate plus Rank/Optimize; Recommend	PDF pp. 16–18, Secs. 3.3 and Conclusions | PDF pp. 17–18, Results and Conclusions | PDF pp. 16–18, Optimization and Conclusions
AIEXP-03223	Kamazani M.A.; Dixit M.K. (2023). Multi-objective optimization of embodied and operational energy and carbon emission of a building envelope. Journal of Cleaner Production. https://doi.org/10.1016/j.jclepro.2023.139510	10.1016/j.jclepro.2023.139510	Transferable	Not Reported	Embodied-carbon database; EnergyPlus; embodied databases; Material/carbon database; Parametric model; genetic algorithm	Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-03238	Li Y.; Mei Y.; Yan X.; Wang S.; Fingrut A. (2025). Digital Framework for Generative Design of Reused Concrete Structure Rubbles. Proceedings of the International Conference on Computer-Aided Architectural Design Research in Asia. https://doi.org/10.52842/conf.caadria.2025.3.457	10.52842/conf.caadria.2025.3.457	Direct	Not Reported	Generated configurations; Generated rubble configurations; Point cloud; rubble inventory	Generative design	Generate plus Rank/Optimize	PDF p. 8, Full text
AIEXP-03242	Gan V.J.L. (2022). BIM-Based Building Geometric Modeling and Automatic Generative Design for Sustainable Offsite Construction. Journal of Construction Engineering and Management. https://doi.org/10.1061/(asce)co.1943-7862.0002369	10.1061/(ASCE)CO.1943-7862.0002369	Transferable	Conceptual	BIM object library; BIM; carbon/cost database; BIM; object library; BIM; parametric model; Carbon-factor database	Generative design; BIM/IFC computational workflow	Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-03246	Sanchez B.; Rausch C.; Haas C.; Saari R. (2019). Multi-objective optimization analysis for selective disassembly planning of buildings. Proceedings of the 36th International Symposium on Automation and Robotics in Construction, ISARC 2019. https://doi.org/10.22260/isarc2019/0018	10.22260/isarc2019/0018	Direct	Not Reported	Cost data; Existing-building inventory; Existing-building inventory; disassembly data	Computational optimization	Generate plus Rank/Optimize	PDF p. 6, Full text
AIEXP-03250	Mirzaei G.; Hammad A.; Homayounfard F. (2025). Design Optimization of Steel Structures with Reused Components Using Generative Design. Proceedings of the International Symposium on Automation and Robotics in Construction. https://doi.org/10.22260/isarc2025/0094	10.22260/ISARC2025/0094	Direct	Not Reported	Parametric model; reused stock; Reused-stock inventory; Reused-stock inventory; structural variables	Evolutionary optimization; Generative design; Computational optimization	Generate plus Rank/Optimize	PDF p. 2, Full text

Record ID	Citation	DOI	Direct/transferable	Context/stage	Source type	AI method	Verified functions	Evidence locator
AIEXP-03264	Kiss B.; Szalay Z. (2023). Sensitivity of buildings' carbon footprint to electricity decarbonization: a life cycle–based multi-objective optimization approach. International Journal of Life Cycle Assessment. https://doi.org/10.1007/s11367-022-02043-y	10.1007/s11367-022-02043-y	Transferable	Not Reported	LCA; energy simulation; Study data source	Computational optimization	Diagnose; Generate plus Rank/Optimize	PDF p. 1, Abstract
AIEXP-03279	Kim S.; Namgung H.; Oh K.; Lee J.; Jeong K.; Hong T.; Kong M. (2025). BIM-IGL: a BIM-based framework for automated conceptual design integrating generative design and life cycle assessment. Energy and Buildings. https://doi.org/10.1016/j.enbuild.2025.116401	10.1016/j.enbuild.2025.116401	Transferable	Conceptual	BIM/parametric model and LCA data; BIM; parametric model	Evolutionary optimization; Generative design; BIM/IFC computational workflow	Generate plus Rank/Optimize	PDF p. 4, Sec. 2.1, Generative design for space layout planning | PDF p. 4, Secs. 2.1-2.2
AIEXP-03283	Dunant C.F.; Drewniok M.P.; Orr J.J.; Allwood J.M. (2021). Good early stage design decisions can halve embodied CO2 and lower structural frames' cost. Structures. https://doi.org/10.1016/j.istruc.2021.04.033	10.1016/j.istruc.2021.04.033	Transferable	Conceptual	Carbon factors; design database; Design database; Design database; generative structural model; Design database; product options; Design database; structural model	Implemented computational workflow	Generate plus Rank/Optimize	PDF p. 6, Sec. 2 | PDF pp. 11–15, Secs. 2.4 and 3 | PDF pp. 6 and 13–15, Secs. 2 and 3 | PDF pp. 13–15, Results | PDF p. 6, Generation algorithm
AIEXP-03292	Zaraza J.; McCabe B.; Duhamel M.; Posen D. (2022). Generative design to reduce embodied GHG emissions of high-rise buildings. Automation in Construction. https://doi.org/10.1016/j.autcon.2022.104274	10.1016/j.autcon.2022.104274	Transferable	Conceptual	BIM; parametric model; BIM; parametric model; simulation output; Embodied-GHG model	Evolutionary optimization; Generative design; Computational optimization	Generate plus Rank/Optimize	PDF p. 52, Sec. 3.4 | PDF pp. 50–53, Secs. 3.3–3.4 | PDF pp. 52–53, Sec. 3.4 | PDF p. 26, Tool inputs
AIEXP-03298	Klejs O.V.; Damgaard M.H.; Jensen M.B. (2026). Designing with Scarcity: A Computational Framework for Embedding Reclaimed Timber in Early Stage Architectural Design. Proceedings of the International Conference on Education and Research in Computer Aided Architectural Design in Europe. https://doi.org/10.52842/conf.ecaade.2026.3.079	10.52842/conf.ecaade.2026.3.079	Direct	Conceptual	Parametric model; reclaimed timber inventory; Reclaimed-timber inventory	Computational optimization	Generate plus Rank/Optimize	PDF p. 6, Full text | PDF p. 3, Full text
AIEXP-03355	Li M.; Zhao S.; Yao S.; Huo Q.; Yuan J.; Li Y. (2025). A performance-responsive generative design framework integrating multi-objective optimization for the layout of residential buildings oriented towards low-carbon emission: A case study of Tianjin in China. Journal of Building Engineering. https://doi.org/10.1016/j.jobe.2025.112805	10.1016/j.jobe.2025.112805	Transferable	Conceptual	Parametric model; Parametric model; neural surrogate; Parametric model; simulation output; Parametric/multi-agent model; Prototype topology	Generative design; Computational optimization	Generate plus Rank/Optimize	PDF p. 1, Abstract | PDF pp. 1 and 3, Abstract; contribution
AIEXP-03807	Jalaei F.; Jrade A.; Nassiri M. (2015). Integrating decision support system (DSS) and building information modeling (BIM) to optimize the selection of sustainable building components. Journal of Information Technology in Construction.		Transferable	Conceptual	BIM; product database; Expert weights; Life-cycle cost data	BIM/IFC computational workflow	Generate plus Rank/Optimize; Recommend	PDF pp. 9 and 17, DSS development and Validation | PDF p. 17, Validation | PDF pp. 11–13, DSS architecture | PDF pp. 9–13, DSS development | PDF pp. 7 and 18, Framework and Validation | PDF pp. 9 and 14–16, DSS development

Table S7. Complete 48 x 6 AI capability matrix in long format.

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
accessibility	Connections and interfaces	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
accessibility	Connections and interfaces	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
accessibility	Connections and interfaces	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
accessibility	Connections and interfaces	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
accessibility	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
accessibility	Connections and interfaces	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
adaptability	Design adaptability and standardization	Generate plus Rank/Optimize	C	No		AIEXP-03172 (Criterion): Adaptability metrics score and rank generated floor plans.	Same-attribute supporting role in the same function; no target/output row
area	Geometry and spatial definition	Extract/Retrieve	A	No	AIEXP-00015; AIEXP-00499		Direct target/output evidence
area	Geometry and spatial definition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
area	Geometry and spatial definition	Predict/Assess	C	No		AIEXP-00034 (Input/Feature): Floor area is an input to demolition-waste prediction.; AIEXP-00450 (Input/Feature): Facade/opening area feeds material-stock and carbon estimation.; AIEXP-01779 (Input/Feature): Schematic area parameters feed embodied-carbon prediction.; AIEXP-01785 (Diagnostic Factor): SHAP identifies floor/land area as schematic carbon drivers.; ... (+2)	Same-attribute supporting role in the same function; no target/output row
area	Geometry and spatial definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
area	Geometry and spatial definition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
area	Geometry and spatial definition	Generate plus Rank/Optimize	C	No		AIEXP-00015 (Constraint): Element area constrains candidate matching.; AIEXP-03283 (Constraint): Floor area and perimeter constrain placement of generated bays.; AIEXP-03355 (Design Variable): Area variables are changed within the generative workflow.	Same-attribute supporting role in the same function; no target/output row
assembly sequence	Component and assembly definition	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
assembly sequence	Component and assembly definition	Infer/Complete	A	No	AIEXP-01773		Direct target/output evidence
assembly sequence	Component and assembly definition	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
assembly sequence	Component and assembly definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
assembly sequence	Component and assembly definition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
assembly sequence	Component and assembly definition	Generate plus Rank/Optimize	A	No	AIEXP-03246; AIEXP-03174		Direct target/output evidence
component count	Quantity and mass	Extract/Retrieve	A	No	AIEXP-00499		Direct target/output evidence
component count	Quantity and mass	Infer/Complete	B	No	AIEXP-00946		Transferable target/output evidence; no Direct row

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
component count	Quantity and mass	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
component count	Quantity and mass	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
component count	Quantity and mass	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
component count	Quantity and mass	Generate plus Rank/Optimize	C	No		AIEXP-03250 (Constraint): Available reused-stock components constrain generation.; AIEXP-03198 (Criterion): Room count evaluates the methods; it is not the generated target by itself.	Same-attribute supporting role in the same function; no target/output row
component identity	Component and assembly definition	Extract/Retrieve	A	No	AIEXP-00183; AIEXP-00189; AIEXP-00216; AIEXP-00474; AIEXP-00469; AIEXP-00499; AIEXP-00536; AIEXP-01043; ... (+1)		Direct target/output evidence
component identity	Component and assembly definition	Infer/Complete	A	Yes	AIEXP-00262; AIEXP-01750; AIEXP-01509; AIEXP-01249; AIEXP-01537; AIEXP-00847; AIEXP-01543; AIEXP-00637; ... (+2)	AIEXP-03196 (Input/Feature): Bim Component Data serves as input/feature evidence and is not treated as the AI target.; AIEXP-01811 (Input/Feature): Building Elements serves as input/feature evidence and is not treated as the AI target.	Direct target/output evidence
component identity	Component and assembly definition	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
component identity	Component and assembly definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
component identity	Component and assembly definition	Recommend	B	Yes	AIEXP-03807	AIEXP-00641 (Input/Feature): Element identity and class are inputs to salvage reasoning.; AIEXP-01773 (Input/Feature): Element identity is an input to sequence and market-link inference.	Transferable target/output evidence; no Direct row
component identity	Component and assembly definition	Generate plus Rank/Optimize	B	No	AIEXP-03217		Transferable target/output evidence; no Direct row
component relationships	Component and assembly definition	Extract/Retrieve	A	No	AIEXP-00189		Direct target/output evidence
component relationships	Component and assembly definition	Infer/Complete	A	No	AIEXP-01773; AIEXP-01509; AIEXP-01537; AIEXP-00800; AIEXP-00946; AIEXP-01769	AIEXP-01249 (Input/Feature): Immediate-neighbor relations provide context for class inference.	Direct target/output evidence

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
component relationships	Component and assembly definition	Predict/Assess	C	No		AIEXP-01864 (Input/Feature): Spatial relationships are GNN input edges, not predicted outputs.	Same-attribute supporting role in the same function; no target/output row
component relationships	Component and assembly definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
component relationships	Component and assembly definition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
component relationships	Component and assembly definition	Generate plus Rank/Optimize	A	Yes	AIEXP-03250; AIEXP-00536; AIEXP-01907; AIEXP-03279; AIEXP-03217	AIEXP-01907 (Constraint): Required adjacencies constrain generated layouts.; AIEXP-03174 (Constraint): Component dependencies constrain sequence generation.; AIEXP-03355 (Constraint): Extracted space relationships constrain generation rather than being the generated target.	Direct target/output evidence
connection type	Connections and interfaces	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
connection type	Connections and interfaces	Infer/Complete	A	No	AIEXP-00495		Direct target/output evidence
connection type	Connections and interfaces	Predict/Assess	C	No		AIEXP-00361 (Input/Feature): Connection variables predict deconstructability.; AIEXP-00533 (Input/Feature): Connection characteristics predict technical reusability.	Same-attribute supporting role in the same function; no target/output row
connection type	Connections and interfaces	Diagnose	C	No		AIEXP-00361 (Diagnostic Factor): Connection variables are contributors, not diagnosis targets.; AIEXP-00533 (Diagnostic Factor): Feature importance tests connection factors as reuse drivers.	Same-attribute supporting role in the same function; no target/output row
connection type	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
connection type	Connections and interfaces	Generate plus Rank/Optimize	C	No		AIEXP-00536 (Constraint): Connection logic constrains generated object placement and fabrication.	Same-attribute supporting role in the same function; no target/output row
deconstruction potential	Disassembly and deconstruction	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
deconstruction potential	Disassembly and deconstruction	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
deconstruction potential	Disassembly and deconstruction	Predict/Assess	A	No	AIEXP-00361		Direct target/output evidence
deconstruction potential	Disassembly and deconstruction	Diagnose	A	No	AIEXP-00361		Direct target/output evidence
deconstruction potential	Disassembly and deconstruction	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
deconstruction potential	Disassembly and deconstruction	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
dimensions	Geometry and spatial definition	Extract/Retrieve	A	No	AIEXP-00015; AIEXP-00189; AIEXP-01743		Direct target/output evidence
dimensions	Geometry and spatial definition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
dimensions	Geometry and spatial definition	Predict/Assess	B	No	AIEXP-01888	AIEXP-00139 (Input/Feature): Envelope and shading dimensions are carbon/comfort model inputs.; AIEXP-00450 (Input/Feature): Thickness and density assumptions feed carbon estimation.	Transferable target/output evidence; no Direct row
dimensions	Geometry and spatial definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
dimensions	Geometry and spatial definition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
dimensions	Geometry and spatial definition	Generate plus Rank/Optimize	B	No	AIEXP-01907; AIEXP-03283	AIEXP-00014 (Constraint): Reclaimed-stock lengths constrain gridshell optimization.; AIEXP-00015 (Constraint): Element length constrains candidate matching.; AIEXP-00536 (Input/Feature): Piece dimensions guide neighborhood matching and placement.; AIEXP-01907 (Design Variable): Room dimensions are varied during layout optimization.; ... (+1)	Transferable target/output evidence; no Direct row
disassembly potential	Disassembly and deconstruction	Extract/Retrieve	A	No	AIEXP-00408		Direct target/output evidence
disassembly potential	Disassembly and deconstruction	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
disassembly potential	Disassembly and deconstruction	Predict/Assess	A	No	AIEXP-00495		Direct target/output evidence
disassembly potential	Disassembly and deconstruction	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
disassembly potential	Disassembly and deconstruction	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
disassembly potential	Disassembly and deconstruction	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
economic value	Supply, take-back, and market conditions	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
economic value	Supply, take-back, and market conditions	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
economic value	Supply, take-back, and market conditions	Predict/Assess	A	Yes	AIEXP-00023; AIEXP-01789; AIEXP-01821	AIEXP-01788 (Input/Feature): Material Cost is used as a feature for embodied-carbon prediction.	Direct target/output evidence
economic value	Supply, take-back, and market conditions	Diagnose	B	Yes	AIEXP-01821		Transferable target/output evidence; no Direct row
economic value	Supply, take-back, and market conditions	Recommend	C	No		AIEXP-01838 (Criterion): Life-cycle cost is retained as a comparative decision criterion.; AIEXP-03807 (Criterion): Life-cycle cost contributes to component selection.	Same-attribute supporting role in the same function; no target/output row
economic value	Supply, take-back, and market conditions	Generate plus Rank/Optimize	C	No		AIEXP-00500 (Criterion): Unit cost is an optimization objective.; AIEXP-03005 (Criterion): Life-cycle cost evaluates building variants.; AIEXP-03246 (Criterion): Deconstruction cost is an optimization objective.; AIEXP-02074 (Criterion): Cost is an optimization criterion.; ... (+5)	Same-attribute supporting role in the same function; no target/output row
embodied carbon	Environmental and resource flows	Extract/Retrieve	A	No	AIEXP-00408		Direct target/output evidence
embodied carbon	Environmental and resource flows	Infer/Complete	C	No		AIEXP-01811 (Input/Feature): Environmental Impact Parameters serves as input/feature evidence and is not treated as the AI target.	Same-attribute supporting role in the same function; no target/output row
embodied carbon	Environmental and resource flows	Predict/Assess	A	Yes	AIEXP-00139; AIEXP-00450; AIEXP-01711; AIEXP-02540; AIEXP-02680; AIEXP-00565; AIEXP-01779; AIEXP-01785; ... (+12)		Direct target/output evidence
embodied carbon	Environmental and resource flows	Diagnose	A	Yes	AIEXP-01711; AIEXP-02540; AIEXP-02680; AIEXP-01785; AIEXP-00350; AIEXP-02655; AIEXP-01872; AIEXP-01821; ... (+1)		Direct target/output evidence

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
embodied carbon	Environmental and resource flows	Recommend	C	No		AIEXP-01838 (Criterion): Embodied-carbon performance contributes to the sustainability objective.; AIEXP-03218 (Criterion): Global-warming and energy indicators evaluate material alternatives.; AIEXP-01796 (Criterion): Embodied carbon is a selection criterion, not the recommended output.	Same-attribute supporting role in the same function; no target/output row
embodied carbon	Environmental and resource flows	Generate plus Rank/Optimize	A	No	AIEXP-02802; AIEXP-02216; AIEXP-01872; AIEXP-03264; AIEXP-03203; AIEXP-03149	AIEXP-00139 (Criterion): Embodied carbon is an optimization objective.; AIEXP-00500 (Criterion): CO2 emissions are an optimization objective.; AIEXP-02074 (Criterion): Carbon footprint is an objective/criterion.; AIEXP-02074 (Diagnostic Factor): SHAP identifies carbon footprint as a driver of the material ranking; it is not the diagnosis target.; ... (+10)	Direct target/output evidence
expert judgment	Scenario and expert inputs	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
expert judgment	Scenario and expert inputs	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
expert judgment	Scenario and expert inputs	Predict/Assess	C	No		AIEXP-01718 (Input/Feature): Human reviewers may accept or override automated compliance outcomes.	Same-attribute supporting role in the same function; no target/output row
expert judgment	Scenario and expert inputs	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
expert judgment	Scenario and expert inputs	Recommend	C	No		AIEXP-03807 (Input/Feature): Expert and practitioner weights define sustainability priorities.	Same-attribute supporting role in the same function; no target/output row
expert judgment	Scenario and expert inputs	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
flexibility	Design adaptability and standardization	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
flexibility	Design adaptability and standardization	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
flexibility	Design adaptability and standardization	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
flexibility	Design adaptability and standardization	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
flexibility	Design adaptability and standardization	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
flexibility	Design adaptability and standardization	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
future value	Supply, take-back, and market conditions	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
geometry	Geometry and spatial definition	Extract/Retrieve	A	No	AIEXP-00015; AIEXP-00183; AIEXP-00471; AIEXP-00499; AIEXP-00536		Direct target/output evidence
geometry	Geometry and spatial definition	Infer/Complete	B	No	AIEXP-01537; AIEXP-00847; AIEXP-01543; AIEXP-00637; AIEXP-01354; AIEXP-00800	AIEXP-01509 (Input/Feature): Geometric calculations support enrichment of missing IFC class and relation information.; AIEXP-01249 (Input/Feature): Geometric features are model inputs; they are not inferred outputs.	Transferable target/output evidence; no Direct row
geometry	Geometry and spatial definition	Predict/Assess	C	No		AIEXP-00023 (Input/Feature): Image geometry and bounding boxes are classifier features.; AIEXP-00183 (Input/Feature): BIM element geometry is used to estimate waste.; AIEXP-01779 (Input/Feature): Schematic geometry feeds embodied-carbon prediction.; AIEXP-01875 (Input/Feature): Parametric geometry feeds the surrogate carbon model.; ... (+4)	Same-attribute supporting role in the same function; no target/output row

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
geometry	Geometry and spatial definition	Diagnose	C	No		AIEXP-03140 (Diagnostic Factor): Geometric variables are tested as waste drivers.; AIEXP-00350 (Diagnostic Factor): SHAP tests the contribution and effect direction of geometric variables.	Same-attribute supporting role in the same function; no target/output row
geometry	Geometry and spatial definition	Recommend	C	No		AIEXP-01773 (Input/Feature): Geometry enables missing-connection inference.	Same-attribute supporting role in the same function; no target/output row
geometry	Geometry and spatial definition	Generate plus Rank/Optimize	A	Yes	AIEXP-00014; AIEXP-00261; AIEXP-03238; AIEXP-03250; AIEXP-03298; AIEXP-00536; AIEXP-03172; AIEXP-03166; ... (+14)	AIEXP-01780 (Design Variable): Window/Volume Variables serves as design variable evidence and is not treated as the AI target.	Direct target/output evidence
LCA Data	Environmental and resource flows	Extract/Retrieve	A	Yes	AIEXP-01636; AIEXP-01743; AIEXP-01656; AIEXP-01724; AIEXP-01731; AIEXP-01687		Direct target/output evidence
LCA Data	Environmental and resource flows	Infer/Complete	A	No	AIEXP-01811	AIEXP-01718 (Input/Feature): Epd/Product Documentation serves as input/feature evidence and is not treated as the AI target.	Direct target/output evidence
LCA Data	Environmental and resource flows	Predict/Assess	A	Yes	AIEXP-01838; AIEXP-01731; AIEXP-02074; AIEXP-01780	AIEXP-01750 (Input/Feature): LCA database values are retrieved inputs to the assessment.; AIEXP-01711 (Input/Feature): EPD fields are model inputs, not extracted outputs in this workflow.	Direct target/output evidence
LCA Data	Environmental and resource flows	Diagnose	A	Yes	AIEXP-01731; AIEXP-01780		Direct target/output evidence
LCA Data	Environmental and resource flows	Recommend	C	No		AIEXP-01902 (Criterion): Lifecycle carbon is the criterion for material selection.; AIEXP-03218 (Criterion): LCA outputs are criteria, not the recommended target.	Same-attribute supporting role in the same function; no target/output row
LCA Data	Environmental and resource flows	Generate plus Rank/Optimize	C	No		AIEXP-03005 (Criterion): LCA impacts evaluate building variants.; AIEXP-03174 (Criterion): Environmental LCA impact is an optimization objective.; AIEXP-03279 (Criterion): LCA results evaluate/rank generated layouts.; AIEXP-03195 (Criterion): LCA indicators evaluate generated facade alternatives.; ... (+3)	Same-attribute supporting role in the same function; no target/output row
location	Geometry and spatial definition	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
location	Geometry and spatial definition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
location	Geometry and spatial definition	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
location	Geometry and spatial definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
location	Geometry and spatial definition	Recommend	C	No		AIEXP-01773 (Input/Feature): Location constrains stakeholder discovery and matching.	Same-attribute supporting role in the same function; no target/output row
location	Geometry and spatial definition	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
maintainability	Service, maintenance, and replacement	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
manufacturer information	Product and provenance metadata	Extract/Retrieve	A	No	AIEXP-00408; AIEXP-01636; AIEXP-01743; AIEXP-01656		Direct target/output evidence
manufacturer information	Product and provenance metadata	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
manufacturer information	Product and provenance metadata	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
manufacturer information	Product and provenance metadata	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
manufacturer information	Product and provenance metadata	Recommend	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
manufacturer information	Product and provenance metadata	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
mass	Quantity and mass	Extract/Retrieve	A	No	AIEXP-00183		Direct target/output evidence
mass	Quantity and mass	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
mass	Quantity and mass	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
mass	Quantity and mass	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
mass	Quantity and mass	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
mass	Quantity and mass	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
material passport	Product and provenance metadata	Extract/Retrieve	A	No	AIEXP-00015; AIEXP-00216; AIEXP-00497		Direct target/output evidence
material passport	Product and provenance metadata	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
material passport	Product and provenance metadata	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
material passport	Product and provenance metadata	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
material passport	Product and provenance metadata	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
material passport	Product and provenance metadata	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
material quantity	Quantity and mass	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
material quantity	Quantity and mass	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
material quantity	Quantity and mass	Predict/Assess	A	No	AIEXP-00110	AIEXP-01785 (Diagnostic Factor): SHAP identifies concrete/reinforcement quantities as later carbon drivers.; AIEXP-01750 (Input/Feature): Element/material quantities feed the GHG calculation.; AIEXP-01872 (Input/Feature): Material Volumes is an input to carbon prediction, not a predicted output.; AIEXP-01788 (Input/Feature): Primary-Material Consumption is used as a feature for embodied-carbon prediction.	Direct target/output evidence
material quantity	Quantity and mass	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
material quantity	Quantity and mass	Recommend	B	Yes	AIEXP-01821		Transferable target/output evidence; no Direct row
material quantity	Quantity and mass	Generate plus Rank/Optimize	C	No		AIEXP-03298 (Constraint): Documented reclaimed stock constrains generated variants.; AIEXP-00536 (Constraint): The available stock constrains the generated solution space.; AIEXP-03279 (Criterion): Material quantity is an evaluation input/criterion, not an AI-generated output.; AIEXP-03166 (Design Variable): Material quantities change with design variables and are not the optimization target themselves.; ... (+1)	Same-attribute supporting role in the same function; no target/output row
material toxicity	Material identity and composition	Extract/Retrieve	A	No	AIEXP-01636; AIEXP-01743; AIEXP-01656		Direct target/output evidence
material toxicity	Material identity and composition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
material toxicity	Material identity and composition	Predict/Assess	A	No	AIEXP-00406; AIEXP-00505		Direct target/output evidence
material toxicity	Material identity and composition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
material toxicity	Material identity and composition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
material toxicity	Material identity and composition	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
material type	Material identity and composition	Extract/Retrieve	A	No	AIEXP-00023; AIEXP-00126; AIEXP-00216; AIEXP-00261; AIEXP-00406; AIEXP-00408; AIEXP-00431; AIEXP-00450; ... (+3)		Direct target/output evidence
material type	Material identity and composition	Infer/Complete	A	Yes	AIEXP-00262; AIEXP-00497; AIEXP-00469; AIEXP-01636; AIEXP-01743; AIEXP-01656; AIEXP-01811; AIEXP-01750; ... (+2)		Direct target/output evidence
material type	Material identity and composition	Predict/Assess	C	No		AIEXP-00034 (Input/Feature): Wall, roof, and floor types are prediction inputs.; AIEXP-00139 (Input/Feature): Material options are prediction inputs.; AIEXP-00183 (Input/Feature): Element/material class is used to estimate waste.; AIEXP-00450 (Input/Feature): Detected facade material class feeds carbon estimation.; ... (+4)	Same-attribute supporting role in the same function; no target/output row
material type	Material identity and composition	Diagnose	C	No		AIEXP-00350 (Diagnostic Factor): Material/structural choices appear as explanatory features rather than diagnosis targets.	Same-attribute supporting role in the same function; no target/output row
material type	Material identity and composition	Recommend	A	Yes	AIEXP-01838; AIEXP-01636; AIEXP-01743; AIEXP-01902; AIEXP-03048; AIEXP-03195; AIEXP-03218; AIEXP-03807; ... (+1)	AIEXP-00641 (Input/Feature): Material associations are inputs to salvage reasoning.; AIEXP-01773 (Input/Feature): Material descriptors are inputs to lexical stakeholder matching.	Direct target/output evidence
material type	Material identity and composition	Generate plus Rank/Optimize	A	Yes	AIEXP-00015; AIEXP-00139; AIEXP-01838; AIEXP-03005; AIEXP-01907; AIEXP-02074; AIEXP-03048; AIEXP-03283; ... (+11)	AIEXP-00261 (Design Variable): Fragment material class is assigned within generated patterns.; AIEXP-00261 (Design Variable): Fragment material class participates in pattern optimization.; AIEXP-03250 (Design Variable): Reused-steel components are assigned in optimization.; AIEXP-00536 (Input/Feature): Material properties accompany scanned piece geometry in the object database.; ... (+7)	Direct target/output evidence
mechanical fastener	Connections and interfaces	Extract/Retrieve	A	No	AIEXP-00406		Direct target/output evidence
mechanical fastener	Connections and interfaces	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
mechanical fastener	Connections and interfaces	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
mechanical fastener	Connections and interfaces	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
mechanical fastener	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
mechanical fastener	Connections and interfaces	Generate plus Rank/Optimize	C	No		AIEXP-03174 (Constraint): Fasteners appear as constrained parts in the disassembly graph.	Same-attribute supporting role in the same function; no target/output row
modularity	Design adaptability and standardization	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
modularity	Design adaptability and standardization	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
modularity	Design adaptability and standardization	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
modularity	Design adaptability and standardization	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
modularity	Design adaptability and standardization	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
modularity	Design adaptability and standardization	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
permanent connection	Connections and interfaces	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
product passport	Product and provenance metadata	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
product passport	Product and provenance metadata	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
product passport	Product and provenance metadata	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
product passport	Product and provenance metadata	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
product passport	Product and provenance metadata	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
product passport	Product and provenance metadata	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
recovery potential	Reuse, recycling, and recovery	Extract/Retrieve	A	Yes	AIEXP-00408; AIEXP-01724		Direct target/output evidence
recovery potential	Reuse, recycling, and recovery	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
recovery potential	Reuse, recycling, and recovery	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
recovery potential	Reuse, recycling, and recovery	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
recovery potential	Reuse, recycling, and recovery	Recommend	A	No	AIEXP-00641	AIEXP-01838 (Criterion): Circularity-focused weighting promotes configurations with improved recovery performance.	Direct target/output evidence
recovery potential	Reuse, recycling, and recovery	Generate plus Rank/Optimize	C	No		AIEXP-03246 (Criterion): Environmental recovery is an optimization objective.	Same-attribute supporting role in the same function; no target/output row
recycled content	Material identity and composition	Extract/Retrieve	A	Yes	AIEXP-00408; AIEXP-01636; AIEXP-01724		Direct target/output evidence
recycled content	Material identity and composition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
recycled content	Material identity and composition	Predict/Assess	C	No		AIEXP-02680 (Input/Feature): Waste-glass content is an input feature, not the predicted target.; AIEXP-02802 (Input/Feature): Waste content is an input to the strength model, not itself predicted.	Same-attribute supporting role in the same function; no target/output row

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
recycled content	Material identity and composition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
recycled content	Material identity and composition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
recycled content	Material identity and composition	Generate plus Rank/Optimize	A	No	AIEXP-00500; AIEXP-02802	AIEXP-02074 (Design Variable): PPO adjusts recycled-content/material formulations.	Direct target/output evidence
recycling potential	Reuse, recycling, and recovery	Extract/Retrieve	A	No	AIEXP-00408; AIEXP-01636		Direct target/output evidence
recycling potential	Reuse, recycling, and recovery	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
recycling potential	Reuse, recycling, and recovery	Predict/Assess	A	No	AIEXP-00023		Direct target/output evidence
recycling potential	Reuse, recycling, and recovery	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
recycling potential	Reuse, recycling, and recovery	Recommend	A	No	AIEXP-00641		Direct target/output evidence
recycling potential	Reuse, recycling, and recovery	Generate plus Rank/Optimize	C	No		AIEXP-02074 (Criterion): Recyclability is an evaluation criterion.; AIEXP-03048 (Criterion): Recyclability is an evaluation criterion; it is not the recommended output.	Same-attribute supporting role in the same function; no target/output row
renewable content	Material identity and composition	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
renewable content	Material identity and composition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
renewable content	Material identity and composition	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
renewable content	Material identity and composition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
renewable content	Material identity and composition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
renewable content	Material identity and composition	Generate plus Rank/Optimize	A	No	AIEXP-02216		Direct target/output evidence
repairability	Service, maintenance, and replacement	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
repairability	Service, maintenance, and replacement	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
repairability	Service, maintenance, and replacement	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
repairability	Service, maintenance, and replacement	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
repairability	Service, maintenance, and replacement	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
repairability	Service, maintenance, and replacement	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
replaceability	Service, maintenance, and replacement	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
reuse potential	Reuse, recycling, and recovery	Extract/Retrieve	A	No	AIEXP-00408		Direct target/output evidence
reuse potential	Reuse, recycling, and recovery	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
reuse potential	Reuse, recycling, and recovery	Predict/Assess	A	No	AIEXP-00023; AIEXP-00262; AIEXP-00533; AIEXP-01859; AIEXP-02103; AIEXP-01718		Direct target/output evidence
reuse potential	Reuse, recycling, and recovery	Diagnose	A	No	AIEXP-00533; AIEXP-01718		Direct target/output evidence
reuse potential	Reuse, recycling, and recovery	Recommend	A	No	AIEXP-00641		Direct target/output evidence

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
reuse potential	Reuse, recycling, and recovery	Generate plus Rank/Optimize	C	No		AIEXP-03238 (Criterion): Reuse feasibility helps select generated rubble configurations.; AIEXP-03298 (Criterion): Stock utilization/reuse rate is an optimization objective.	Same-attribute supporting role in the same function; no target/output row
reversible connection	Connections and interfaces	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
reversible connection	Connections and interfaces	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
reversible connection	Connections and interfaces	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
reversible connection	Connections and interfaces	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
reversible connection	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
reversible connection	Connections and interfaces	Generate plus Rank/Optimize	C	No		AIEXP-00536 (Criterion): Reversibility is a stated joining objective rather than an AI-generated target.	Same-attribute supporting role in the same function; no target/output row
scenario assumptions	Scenario and expert inputs	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
scenario assumptions	Scenario and expert inputs	Infer/Complete	C	No		AIEXP-01780 (Input/Feature): Climate Scenario serves as input/feature evidence and is not treated as the AI target.	Same-attribute supporting role in the same function; no target/output row
scenario assumptions	Scenario and expert inputs	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
scenario assumptions	Scenario and expert inputs	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
scenario assumptions	Scenario and expert inputs	Recommend	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
scenario assumptions	Scenario and expert inputs	Generate plus Rank/Optimize	C	No		AIEXP-03172 (Input/Feature): Future-living scenarios define resilience-sensitive rooms used in evaluation.; AIEXP-03264 (Criterion): Electricity Mix Scenario serves as criterion evidence and is not treated as the AI target.; AIEXP-03203 (Design Variable): Refurbishment Scenarios serves as design variable evidence and is not treated as the AI target.; AIEXP-01718 (Criterion): Future Reuse Conditions serves as criterion evidence and is not treated as the AI target.	Same-attribute supporting role in the same function; no target/output row
separability	Connections and interfaces	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
separability	Connections and interfaces	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
separability	Connections and interfaces	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
separability	Connections and interfaces	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
separability	Connections and interfaces	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
separability	Connections and interfaces	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
service life	Service, maintenance, and replacement	Extract/Retrieve	A	No	AIEXP-00408; AIEXP-01636		Direct target/output evidence
service life	Service, maintenance, and replacement	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
service life	Service, maintenance, and replacement	Predict/Assess	A	No	AIEXP-00262		Direct target/output evidence
service life	Service, maintenance, and replacement	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
service life	Service, maintenance, and replacement	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
service life	Service, maintenance, and replacement	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
standardization	Design adaptability and standardization	Extract/Retrieve	D	No			No target/output evidence or qualifying same-function supporting role
standardization	Design adaptability and standardization	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
standardization	Design adaptability and standardization	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
standardization	Design adaptability and standardization	Diagnose	C	No		AIEXP-02655 (Diagnostic Factor): Standardization is diagnosed as a driver of carbon, not itself diagnosed.	Same-attribute supporting role in the same function; no target/output row
standardization	Design adaptability and standardization	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
standardization	Design adaptability and standardization	Generate plus Rank/Optimize	A	Yes	AIEXP-03166		Direct target/output evidence
supply-chain information	Supply, take-back, and market conditions	Extract/Retrieve	A	No	AIEXP-00641; AIEXP-01773		Direct target/output evidence
supply-chain information	Supply, take-back, and market conditions	Infer/Complete	A	No	AIEXP-00641		Direct target/output evidence
supply-chain information	Supply, take-back, and market conditions	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
supply-chain information	Supply, take-back, and market conditions	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
supply-chain information	Supply, take-back, and market conditions	Recommend	A	No	AIEXP-01773	AIEXP-00641 (Criterion): Policy and facility constraints determine disposition recommendations.	Direct target/output evidence
supply-chain information	Supply, take-back, and market conditions	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
take-back information	Supply, take-back, and market conditions	Extract/Retrieve	A	No	AIEXP-01773		Direct target/output evidence
take-back information	Supply, take-back, and market conditions	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
take-back information	Supply, take-back, and market conditions	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
take-back information	Supply, take-back, and market conditions	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
take-back information	Supply, take-back, and market conditions	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
take-back information	Supply, take-back, and market conditions	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
transport	Environmental and resource flows	Extract/Retrieve	A	No	AIEXP-00408		Direct target/output evidence
transport	Environmental and resource flows	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
transport	Environmental and resource flows	Predict/Assess	C	No		AIEXP-01872 (Input/Feature): Transportation is an input to carbon prediction, not a predicted output.	Same-attribute supporting role in the same function; no target/output row
transport	Environmental and resource flows	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
transport	Environmental and resource flows	Recommend	B	Yes	AIEXP-01821		Transferable target/output evidence; no Direct row
transport	Environmental and resource flows	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
virgin material	Material identity and composition	Extract/Retrieve	A	No	AIEXP-01636		Direct target/output evidence
virgin material	Material identity and composition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
virgin material	Material identity and composition	Predict/Assess	D	No			No target/output evidence or qualifying same-function supporting role
virgin material	Material identity and composition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
virgin material	Material identity and composition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
virgin material	Material identity and composition	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
volume	Geometry and spatial definition	Extract/Retrieve	A	No	AIEXP-00183; AIEXP-00499		Direct target/output evidence
volume	Geometry and spatial definition	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role

Attribute	Family	Function	Grade	Early design	Evidence paper IDs	Supporting role / C basis	Audit note
volume	Geometry and spatial definition	Predict/Assess	A	Yes	AIEXP-00246; AIEXP-01871	AIEXP-02655 (Input/Feature): Component volume is an input feature, not the predicted target.	Direct target/output evidence
volume	Geometry and spatial definition	Diagnose	D	No			No target/output evidence or qualifying same-function supporting role
volume	Geometry and spatial definition	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
volume	Geometry and spatial definition	Generate plus Rank/Optimize	D	No			No target/output evidence or qualifying same-function supporting role
waste	Environmental and resource flows	Extract/Retrieve	A	No	AIEXP-00038; AIEXP-00408		Direct target/output evidence
waste	Environmental and resource flows	Infer/Complete	D	No			No target/output evidence or qualifying same-function supporting role
waste	Environmental and resource flows	Predict/Assess	A	Yes	AIEXP-00034; AIEXP-00069; AIEXP-00183; AIEXP-00246; AIEXP-00498; AIEXP-02243; AIEXP-03140; AIEXP-02074		Direct target/output evidence
waste	Environmental and resource flows	Diagnose	A	Yes	AIEXP-03140		Direct target/output evidence
waste	Environmental and resource flows	Recommend	D	No			No target/output evidence or qualifying same-function supporting role
waste	Environmental and resource flows	Generate plus Rank/Optimize	C	No		AIEXP-00536 (Criterion): Material waste is one optimization criterion.; AIEXP-03174 (Criterion): Discarded parts contribute to the environmental objective.	Same-attribute supporting role in the same function; no target/output row

Supplementary figures

Information-source distribution across all 48 normalized circularity attributes

Unique primary-review records with each locked source-group code; source groups are non-exclusive.

Normalized attribute	Drawing CAD	BIM IFC	Specification BoQ	Product metadata	EPD/LCA database	Existing-building inventory	Scenario judgment	Not reported
Component and assembly definition								
Assembly Sequence	0	0	0	0	0	0	0	43
Component Identity	0	0	0	0	0	0	0	45
Component Relationships	0	0	0	0	0	0	0	476
Connections and interfaces								
Accessibility	3	24	1	11	7	4	5	269
Connection Type	0	0	0	0	0	0	0	363
Mechanical Fastener	0	0	0	0	0	0	0	37
Permanent Connection	0	0	0	0	0	0	0	35
Reversible Connection	0	0	0	0	0	0	0	119
Separability	0	2	0	2	0	0	0	23
Design adaptability and standardization								
Adaptability	1	30	3	15	12	4	5	197
Flexibility	6	26	7	19	6	0	8	218
Modularity	0	0	0	0	0	0	0	304
Standardization	0	0	0	0	0	0	0	449
Disassembly and deconstruction								
Deconstruction Potential	2	48	8	20	4	7	6	231
Disassembly Potential	3	32	3	21	9	3	4	137
Environmental and resource flows								
Embodied Carbon	1	66	18	33	45	8	7	232
Life-cycle Assessment Data	3	34	12	6	58	4	6	302
Transport	0	4	2	2	8	0	5	221
Waste	4	117	19	42	27	14	16	419
Geometry and spatial definition								
Area	0	0	0	0	0	0	0	482
Dimensions	0	0	0	0	0	0	0	532
Geometry	0	0	0	0	0	0	0	370
Location	0	0	0	0	0	0	0	417
Volume	0	0	0	0	0	0	0	372
Material identity and composition								
Material Toxicity	0	12	2	5	5	5	3	129
Material Type	0	0	0	0	0	0	0	116
Recycled Content	0	6	2	6	3	0	4	24
Renewable Content	1	1	0	0	0	0	0	51
Virgin Material	0	2	1	3	2	0	1	21
Product and provenance metadata								
Manufacturer Information	0	3	1	16	3	1	0	149
Material Passport	0	36	3	109	11	3	3	13
Product Passport	1	12	2	58	4	1	1	3
Quantity and mass								
Component Count	0	0	0	0	0	0	0	103
Mass	0	0	0	0	0	0	0	414
Material Quantity	0	0	0	0	0	0	0	247
Reuse, recycling, and recovery								
Recovery Potential	1	15	2	12	8	3	1	56
Recycling Potential	2	39	6	43	17	6	6	188
Reuse Potential	6	50	5	42	15	10	10	461
Scenario and expert inputs								
Expert Judgment	0	4	1	1	1	0	18	230
Scenario Assumptions	0	9	1	6	4	2	51	225
Service, maintenance, and replacement								
Maintainability	0	0	0	1	1	0	0	17
Repairability	0	1	2	2	0	0	0	17
Replaceability	0	0	0	0	0	0	0	8
Service Life	0	22	1	16	9	2	7	223
Supply, take-back, and market conditions								
Economic Value	0	10	1	10	3	0	1	534
Future Value	0	1	0	1	0	0	0	21
Supply-chain Information	0	2	1	4	0	0	1	273
Take-back Information	0	1	0	2	1	1	0	15

Shading log-scaled for legibility; printed values are exact record counts

Not Reported includes relationships without a recorded broad source-group code; no source category was inferred from drawing-extractability fields.

A record may contribute to more than one source category for the same attribute when multiple sources were explicitly coded.

Figure S1. Information-source distribution across all 48 normalized circularity attributes. Cell values are counts of unique primary-review records with each locked broad source-group code. Source groups are non-exclusive. Not Reported includes relationships without a recorded broad source-group code; source categories were not inferred from drawing-extractability fields.

Normalized 48-attribute circularity-information vocabulary

Frequency is the number of unique circularity papers requiring the attribute; wording definitions and representative Record IDs remain traceable

Family	Normalized attribute	n	Readiness	Information meaning	Representative source records
Component and assembly definition	**Assembly Sequence**	43	Core	Order in which components are assembled, removed, or disassembled.	R00016; R00018; R00152; R00254; R00279; R00392; …
	Component Identity	45	Core	Generic or specific identification of a building component or element type.	R00024; R00028; R00055; R00056; R00096; R00121; …
	Component Relationships	476	Core	Adjacency, containment, support, connectivity, or assembly relationships among componen…	R00005; R00012; R00017; R00018; R00021; R00024; …
Connections and interfaces	**Accessibility**	316	Unresolved	Physical or visual access needed to inspect, service, remove, or separate a component.	R00002; R00005; R00006; R00007; R00017; R00024; …
	Connection Type	363	Core	Generic or detailed means by which components are joined.	R00002; R00005; R00006; R00016; R00018; R00021; …
	Mechanical Fastener	37	Core	Presence and type of bolts, screws, clips, or other mechanical fasteners.	R00208; R00302; R00324; R00359; R00363; R00378; …
	Permanent Connection	35	Core	Presence of irreversible or difficult-to-separate joining methods.	R00125; R00202; R00206; R00219; R00302; R00429; …
	Reversible Connection	119	Core	Connection that can be undone without unacceptable damage to joined parts.	R00006; R00018; R00062; R00066; R00075; R00078; …
	Separability	27	Unresolved	Ability to separate materials or components without contamination or destructive loss.	R00047; R00116; R00212; R00229; R00253; R00414; …
Design adaptability and standardization	**Adaptability**	251	Unresolved	Capacity of a building, system, or component to accommodate future change.	R00005; R00006; R00007; R00012; R00024; R00031; …
	Flexibility	281	Unresolved	Capacity to support alternative layouts, uses, or configurations.	R00006; R00017; R00018; R00024; R00030; R00031; …
	Modularity	304	Core	Use of coordinated modules, grids, kits, or repeatable dimensional units.	R00002; R00005; R00006; R00007; R00012; R00017; …
	Standardization	449	Core	Use of standard sizes, interfaces, products, or repeated component rules.	R00002; R00006; R00007; R00009; R00012; R00016; …
Disassembly and deconstruction	**Deconstruction Potential**	312	Unresolved	Potential for planned dismantling or selective deconstruction rather than destructive d…	R00002; R00006; R00012; R00016; R00017; R00040; …
	Disassembly Potential	193	Unresolved	Potential to separate components without destructive loss of function or value.	R00006; R00012; R00016; R00056; R00058; R00062; …
Environmental and resource flows	**Embodied Carbon**	362	Unresolved	Embodied greenhouse-gas impact associated with materials, products, or assemblies.	R00002; R00005; R00006; R00016; R00024; R00040; …
	Life-Cycle Assessment Data	377	Core	Inventory, characterization factors, or other data used in life-cycle assessment.	R00002; R00006; R00007; R00009; R00016; R00017; …
	Transport	234	Unresolved	Distance, mode, route, or logistics information for material and product movement.	R00005; R00018; R00040; R00047; R00055; R00056; …
	Waste	610	Unresolved	Expected or actual waste quantity, type, or rate across lifecycle activities.	R00002; R00005; R00006; R00007; R00012; R00016; …
Geometry and spatial definition	**Area**	482	Core	Plan, surface, gross, net, or component area used in a circularity calculation.	R00006; R00009; R00012; R00018; R00024; R00028; …
	Dimensions	532	Core	Linear dimensions, thicknesses, spans, or sizes used in a circularity calculation.	R00005; R00006; R00007; R00012; R00016; R00017; …
	Geometry	370	Core	Spatial form, shape, topology, or geometric representation of building elements.	R00006; R00012; R00016; R00017; R00018; R00021; …
	Location	417	Core	Site, spatial position, orientation, or element location relevant to an assessment.	R00012; R00016; R00017; R00018; R00021; R00024; …
	Volume	372	Core	Three-dimensional quantity of space, material, or component geometry.	R00002; R00006; R00012; R00017; R00018; R00021; …
Material identity and composition	**Material Toxicity**	153	Unresolved	Hazardous substance, toxicity, or health-related material property.	R00006; R00030; R00056; R00066; R00096; R00101; …
	Material Type	116	Core	Generic material family, specific material, or product-level material identity.	R00002; R00024; R00056; R00073; R00075; R00096; …
	Recycled Content	38	Unresolved	Share of recycled feedstock in a material or product.	R00097; R00108; R00139; R00151; R00196; R00209; …
	Renewable Content	53	Unresolved	Share of renewable or bio-based content in a material or product.	R00005; R00062; R00097; R00120; R00151; R00155; …
	Virgin Material	27	Unresolved	Share or quantity of virgin feedstock in a material or product.	R00130; R00151; R00251; R00332; R00336; R00348; …
Product and provenance metadata	**Manufacturer Information**	165	Unresolved	Manufacturer-specific product identity, properties, documentation, and service informat…	R00002; R00012; R00017; R00021; R00035; R00040; …
	Material Passport	122	Unresolved	Structured material-level identity, composition, provenance, and circularity record.	R00002; R00017; R00035; R00041; R00043; R00047; …
	Product Passport	61	Unresolved	Structured product-level identity, properties, provenance, and circularity record.	R00002; R00030; R00043; R00051; R00056; R00084; …
Quantity and mass	**Component Count**	103	Core	Number of components or repeated elements required by an assessment.	R00006; R00228; R00332; R00343; R00054; R00065; …
	Mass	414	Core	Material or component mass used in material-flow or impact calculations.	R00006; R00017; R00018; R00024; R00028; R00030; …
	Material Quantity	247	Core	Quantity of a material by volume, mass, area, or another measure.	R00007; R00009; R00017; R00024; R00040; R00043; …
Reuse, recycling, and recovery	**Recovery Potential**	88	Unresolved	Potential for recovery through reuse, remanufacture, recycling, or other pathways.	R00007; R00066; R00117; R00121; R00130; R00138; …
	Recycling Potential	288	Unresolved	Potential for a material or component to be recycled under specified conditions.	R00002; R00006; R00007; R00009; R00017; R00035; …
	Reuse Potential	562	Unresolved	Potential for a component or material to be reused in the same or another application.	R00002; R00005; R00006; R00007; R00009; R00012; …
Scenario and expert inputs	**Expert Judgment**	250	Unresolved	Human interpretation, weighting, or decision input required by an assessment.	R00002; R00006; R00007; R00009; R00031; R00035; …
	Scenario Assumptions	280	Unresolved	Explicit assumptions describing future use, lifetime, recovery, market, or system condi…	R00012; R00032; R00035; R00040; R00048; R00055; …
Service, maintenance, and replacement	**Maintainability**	19	Unresolved	Ease and feasibility of inspection, servicing, and maintenance.	R00181; R00240; R00436; R00577; R00682; R00703; …
	Repairability	20	Unresolved	Ease and feasibility of repairing a component or system.	R00002; R00104; R00181; R00196; R00269; R00279; …
	Replaceability	8	Unresolved	Ease and feasibility of replacing a component without disproportionate disturbance.	R00125; R00139; R00321; R00563; R01043; R01108; …
	Service Life	273	Unresolved	Expected or actual period over which a building, component, or material remains functio…	R00006; R00009; R00012; R00035; R00040; R00043; …
Supply, take-back, and market conditions	**Economic Value**	557	Unresolved	Current cost, residual value, or economic quantity used in a circularity assessment.	R00002; R00005; R00006; R00007; R00012; R00016; …
	Future Value	22	Unresolved	Estimated value of a material or component at a future recovery or reuse time.	R00016; R00043; R00142; R00173; R00236; R00302; …
	Supply-Chain Information	279	Unresolved	Supplier availability, sourcing, provenance, logistics, and procurement information.	R00002; R00006; R00007; R00017; R00021; R00030; …
	Take-Back Information	20	Unresolved	Supplier, manufacturer, or contractual arrangements for accepting products or materials…	R00166; R00279; R00288; R00413; R00516; R00600; …

Figure S2. Complete normalized 48-attribute circularity-information vocabulary by navigation family. Frequency is the number of unique circularity papers requiring the attribute. Readiness indicates the final source-grounded Core/non-Core classification.

Complete 48 × 6 AI-capability matrix

Appendix view. Matrix grades describe exact-attribute target/output capability.

Attribute	Extract	Infer	Predict	Diagnose	Recommend	Generate / Rank
Connections and interfaces						
Accessibility	D	D	D	D	D	D
● **Connection Type**	D	A	C	C	D	C
● **Mechanical Fastener**	A	D	D	D	D	C
● **Permanent Connection**	D	D	D	D	D	D
● **Reversible Connection**	D	D	D	D	D	C
Separability	D	D	D	D	D	D
Design adaptability and standardization						
Adaptability	D	D	D	D	D	C
Flexibility	D	D	D	D	D	D
● **Modularity**	D	D	D	D	D	D
● **Standardization**	D	D	D	C	D	A ●
Geometry and spatial definition						
● **Area**	A	D	C	D	D	C
● **Dimensions**	A	D	B	D	D	B
● **Geometry**	A	B	C	C	C	A ●
● **Location**	D	D	D	D	C	D
● **Volume**	A	D	A ●	D	D	D
Component and assembly definition						
● **Assembly Sequence**	D	A	D	D	D	A
● **Component Identity**	A	A ●	D	D	B ●	B
● **Component Relationships**	A	A	C	D	D	A ●
Quantity and mass						
● **Component Count**	A	B	D	D	D	C
● **Mass**	A	D	D	D	D	D
● **Material Quantity**	D	D	A	D	B ●	C
Disassembly and deconstruction						
Deconstruction Potential	D	D	A	A	D	D
Disassembly Potential	A	D	A	D	D	D
Supply, take-back, and market conditions						
Economic Value	D	D	A ●	B ●	C	C
Future Value	D	D	D	D	D	D
Supply-Chain Information	A	A	D	D	A	D
Take-Back Information	A	D	D	D	D	D
Environmental and resource flows						
Embodied Carbon	A	C	A ●	A ●	C	A
● **Life-Cycle Assessment Data**	A ●	A	A ●	A ●	C	C
Transport	A	D	C	D	B ●	D
Waste	A	D	A ●	A ●	D	C
Scenario and expert inputs						
Expert Judgment	D	D	C	D	C	D
Scenario Assumptions	D	C	D	D	D	C
Service, maintenance, and replacement						
Maintainability	D	D	D	D	D	D
Repairability	D	D	D	D	D	D
Replaceability	D	D	D	D	D	D
Service Life	A	D	A	D	D	D
Product and provenance metadata						
Manufacturer Information	A	D	D	D	D	D
Material Passport	A	D	D	D	D	D
Product Passport	D	D	D	D	D	D
Material identity and composition						
Material Toxicity	A	D	A	D	D	D
● **Material Type**	A	A ●	C	C	A ●	A ●
Recycled Content	A ●	D	C	D	D	A
Renewable Content	D	D	D	D	D	A
Virgin Material	A	D	D	D	D	D
Reuse, recycling, and recovery						
Recovery Potential	A ●	D	D	D	A	C
Recycling Potential	A	D	A	D	A	C
Reuse Potential	A	D	A	A	A	C

A direct · B transferable · C plausible · D no qualifying evidence · ● Core attribute · ● early-design cell

Information readiness remains independent from demonstrated AI capability.

Figure S3. Complete 48 x 6 AI capability matrix. A denotes direct circularity target/output evidence, B transferable building-design target/output evidence, C a verified supporting-role plausibility basis, and D no qualifying evidence in the locked corpus. Early-design markers describe study context and do not alter information-readiness classification.